

Maine Aquatic Barrier Prioritization Tool

Version 2.0 – April 2026



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1 Introduction

This report provides an overview of the new Maine Aquatic Barrier Prioritization Tool published by The Nature Conservancy (TNC) in Maine in April 2026. It explores the theory and analyses behind the tool and provides a brief introduction to using and navigating the tool. More detailed instructions on using the tool along with a detailed Data Sources (Appendix III) and Glossary of Metric Definitions (Appendix IV) are provided within the tool itself.

The Maine Aquatic Barrier Prioritization Tool (ABPT) is a prioritization of dams and road-stream crossings that assesses each barrier for the potential ecological benefit that could be realized by removing or upgrading the barrier. With the stream restoration community as its target audience, the tool is designed to help users understand the physical and ecological context of each barrier to help inform an understanding of which barriers to focus resources to help improve stream connectivity for species of interest. This information must be used in concert with other relevant data (social, economic, feasibility) to identify actionable projects.

This tool enables the user to view barriers prioritized by five fish prioritization scenarios run at the statewide, HUC8, and HUC10 watershed scales. In addition, the user can filter among crossing data, view extensive details associated with each crossing and dam, view the GIS datasets that contributed to the prioritization scenarios, and export prioritized and filtered subsets of the crossing data.

The project is an update of the tool and analysis originally published by Martin in 2019, which was based largely on the Penobscot Habitat Blueprint barrier prioritization that was developed by The Nature Conservancy in Maine in 2018. The tool needed to be migrated to a new platform, presenting an opportunity to perform additional work to update the input data, analyses, and user interface and functions (see Section 1.1). Importantly, the original metrics and weighting of all five prioritization scenarios were retained. Modifications outside of the metrics and weighting were made to the Atlantic salmon (*Salmo salar*) scenario, described in Section 4.3.3.

The new Maine Aquatic Barrier Prioritization Tool can be accessed through the TNC Maine Mapping Portal (<https://www.maps.tnc.org/maine/>). The portal hosts a variety of apps that are designed to support conservation planning in Maine.

Please direct questions and feedback about the tool to nature@tnc.org.

1.1 What's New in 2026 – Significant Changes from Previous Edition

- **Data updates:** Major updates and improvements to the barrier, habitat, and landscape data that drives the prioritization metrics and scenarios. See Data Sources Appendix (Appendix III) for details.
- **More advanced 'Help' section:** A redesigned, detailed Help section that gives easy access to both the mechanics of using the tool and all the data and analyses behind it.

- **Prioritization Data and Reporting:** A new data structure 'behind-the-scenes' that gives easier access to multiple prioritization scenarios and a new prioritization tiering scheme that shows 5 results tiers (Tier 1 being the Top 10%), giving the user access to greater scoring resolution.
- **Atlantic salmon prioritization scenario:** Modified Atlantic salmon prioritization scenarios based on extensive consultation with Maine DMR, USFWS, and NOAA experts. This updated version provides greater nuance to which barriers can be included in this scenario, based on the latest thinking and habitat priorities for the species.
- **Multi-species filtering:** New data structure and app design allows the user to consider multiple species prioritizations simultaneously. This allows the user to see which set of barriers are a high priority for a suite of species with different habitat needs. This feature was added to encourage a more holistic, multi-species approach.
- **Increased ease and options for filtering:** A new filtering interface of grouped filters that allow for easy filtering across different types of data using a simple interface. This allows the user to further hone the results of the prioritization. Importantly – filtering prioritized results does not change the actual prioritization, it refines the set of crossings that the user sees.
- **Increased capacity to see barrier specific data, including photos:** A new 'Details' tab that gives the user access to extensive barrier-specific information, including basic location information, date last surveyed, all metrics that drive species prioritization scenarios, species prioritization results, barrier photos, and more.
- **Network Tracing visualization:** A new feature that allows the user to click on a barrier in the map and see its full upstream and downstream network highlighted. This allows the user to see a barrier's place in the overall network and get a feel for its place in the landscape.
- **Data Export (advanced capabilities):** A redesigned Data Export function that allows the user to export their own custom-filtered dataset and choose the categories of data attributes they want to download.
- **Internal TNC Version with Login:** A new user-level login for TNC users that gives access to barrier data that is not shared on the public web application. All data, including photos, are held on an internal TNC server and access is granted by request.

1.2 Uses of the Results

The results in the Maine Aquatic Barrier Prioritization Tool can be used in several different ways to inform and support on-the-ground efforts to restore aquatic connectivity. As discussed by McKay et al (2020) barrier prioritizations of this type include several strengths including their minimally prescriptive nature and allowing for the incorporation of additional site-specific knowledge. Thus, these results are best seen as a screening tool that can be layered with additional information on feasibility, opportunity, economic, and other factors that are not available at the statewide scale.

- **Project Selection, at numerous geographic scales:** A primary use is to help managers direct limited resources to projects that can have the greatest benefit; to help them move from a purely opportunistic approach to an ecological benefits approach (recognizing that opportunity is among other non-ecological factors that continue to play a key role in project selection).

- **Improve Understanding of Current Conditions:** Prioritization results can be used to help direct managers to investigate previously unvisited crossings and dams to assess them for potential passage projects. In some cases, this may reveal errors in the source data while in other cases it may direct attention to potential projects that hadn't been considered previously.
- **Database of Ecologically Relevant Metrics:** Prioritization aside, the results form a database of metrics that can be used to investigate many aspects of aquatic connectivity on a barrier-by-barrier basis or other supplementary analyses.
- **Funding:** The prioritized results can be used both by managers seeking funding for a potential project as well as by funders looking for information to inform or support funding allocation decisions.
- **Communication:** Results can be used to communicate the value of a given project to the local community, elected officials, or others with an interest in aquatic connectivity issues.

1.3 Caveats & Limitations

As with any desktop GIS analysis, there are several caveats and limitations that are important to bear in mind when considering the results and data produced by this project.

First and foremost among them, the results are *not intended to be a hit list* of dams or road-stream crossings for removal. There are many cases where the benefits provided by a given barrier outweigh the ecological benefits of removing it, although other types of passage projects can be considered when removal is not the best option.

Next, this project, by design, only considers ecological factors. It does *not include social, economic, or feasibility factors*, largely due to the fact that this information is difficult or impossible to capture through state-scale GIS datasets. These factors can be layered onto the prioritization results through a subsequent site-scale analysis and partner conversations.

Results produced for this project are intended to be *screening-level* information that can *help* inform on-the-ground decision making, using the best available regional data. They are not a replacement for site-specific knowledge and field work.

Finally, it is important to note that any aquatic connectivity project will have ecological benefits and if an opportunity arises it should not be rejected solely on the grounds that it does not rank highly in one of the upper tiers in a scenario. Ultimately, whether the benefits provided by a given passage project justify the costs is a decision that rests with managers using all of the best information at their disposal.

We hope that this project will be a useful and important tool in the aquatic connectivity toolkit to help provide context for barriers and their impacts on aquatic connectivity across the state.



2 Background and Approach

2.1 Background

The anthropogenic fragmentation of river habitats through dams and poorly designed road-stream crossings is one of the primary threats to aquatic species in the United States (Collier et al. 1997, Graf 1999). The impact of fragmentation on aquatic species generally involves loss of access to quality habitat for one or more life stages of a species. For example, dams and impassable culverts limit the ability of anadromous fish species to reach preferred spawning habitats and can prevent Eastern Brook trout (*Salvelinus fontinalis*) populations from reaching thermal refuges.

Some dams provide valuable services to society including low-carbon electricity, flood control, and irrigation. Many more dams, however, no longer provide the services for which they were designed (e.g. old mill dams) or are structurally failing due to age or design. However, these dams still create barriers to aquatic organism passage. Multiple agencies and organizations in Maine are committed to restoring habitat for anadromous and resident fish species including the Endangered Species Act-listed



Figure 2-1: Many old dams are relics of the industrial history of the region

Atlantic salmon, alewife (*Alosa pseudoharengus*) and Eastern Brook trout. Connectivity restoration projects including dam removals, culvert upgrades, and the installation of fish passage facilities have yielded ecological benefits such as increased anadromous fish runs, improved habitat quality for Eastern Brook trout, and expanded mussel populations. These projects have been spearheaded by state agencies, federal agencies, municipalities, NGOs, and private

corporations – often working in partnership. Substantial funding for these projects has come from the federal government, including the National Oceanic and Atmospheric Administration (NOAA), United States Fish and Wildlife Service (USFWS), and Natural Resources Conservation Service (NRCS); funding has also come from state and private sources. All funding sources have been impacted by recent fiscal instability and federal funding for connectivity restoration is subject to significant budget tightening that requires increased accountability for ecological outcomes.

To many working in the field of aquatic resource management it is apparent that given likely future constraints on availability of funds and staffing, it will be critical to be more strategic about investments in connectivity restoration projects. One approach to strategic investment is to assess the likely ecological “return on investment” associated with connectivity restoration.

The sections that follow detail the data, methods, results, and tools developed for the new Maine ABPT.

2.2 Approach

2.2.1 Workgroup

The first version of the ABPT project was structured around a project Workgroup, composed of members from federal & state agencies, tribal governments, NGOs, and academia (see Appendix II). Meeting both in-person and virtually, the Workgroup was involved in several key aspects of the project including data acquisition & review, key decision making, and draft result review. This collaborative workgroup approach built upon TNC's successful experience working with partners to complete the Northeast Aquatic Connectivity (Martin & Apse 2011) and subsequent projects. In addition to providing input at key points in the project, the Workgroup members form a core user base, who are active in aquatic connectivity restoration with a direct interest in the results.

Central among the key decisions made by the Workgroup was to define the objectives of the prioritization. That is, 1) what is the objective - what are we prioritizing for the benefit of? and 2) what aspects of a barrier or its location would make its removal help achieve the intended objective? This process of selecting objectives and metrics that would be used to evaluate the barriers was both a collaborative and subjective process. The Workgroup selected five prioritization objectives: Alewife, coastal anadromous fish, inland Eastern Brook trout, Atlantic salmon, and American shad (*Alosa sapidissima*) / Blueback herring (*Alosa aestivalis*). Different metrics were used to create five separate prioritization scenarios for these five objectives. Further, each scenario was stratified by HUC8 and HUC10 watershed to produce relative results for each watershed as well as the statewide extent.



Figure 2-2: Project Extent (State of Maine)

2.2.2 Project Extent & Input Barriers

The project extent covers the state of Maine. Within this extent, project results are available for all dams and public road-stream crossings. In addition to the crossings included in the results, additional barriers were included in the input data as needed to generate metrics. For example, in order to calculate the number of downstream barriers for each barrier in the Wolastoq-St John River basin, it was necessary to include Canadian barriers since the river drains into New Brunswick. After Canadian barriers were used to generate metrics, they were removed from the analysis. More information on the barriers used in the analysis can be found in Section 3.3.

3 Data Collection and Preparation

Spatial data for the project were gathered from multiple data sources and processed in a Geographic Information System (GIS) to generate descriptive metrics for each dam and road-stream crossing. The core datasets included river hydrography, dams, road-stream crossings and natural barriers (e.g. waterfalls). Additional datasets were brought in as needed to generate the metrics by which each barrier was evaluated (see Figure 4-1). These datasets include but are not limited to: land cover data, multiple Atlantic salmon data layers, and Eastern Brook trout data. A complete list of data used in the project can be found in Appendix III. A further description of the core datasets, fundamental metric concepts, and data preparation steps follows.

3.1 Definitions

Several terms are used throughout the discussion of data and metrics. The sections below detail some important terms for understanding the data and how metrics were calculated.

3.1.1 Functional River Networks

A barrier's functional river network, also referred to as its connected river network or simply its network, is defined by those stream reaches that are accessible to a hypothetical fish within that network. A given barrier's **functional river network** is bounded by other barriers, headwaters, or the river mouth, as is illustrated in Figure 3-1. A barrier's **total functional river network** is simply the combination of its upstream and downstream functional river networks. The total functional network represents the total river network within which a fish could theoretically swim within if that particular barrier was removed or made passable.

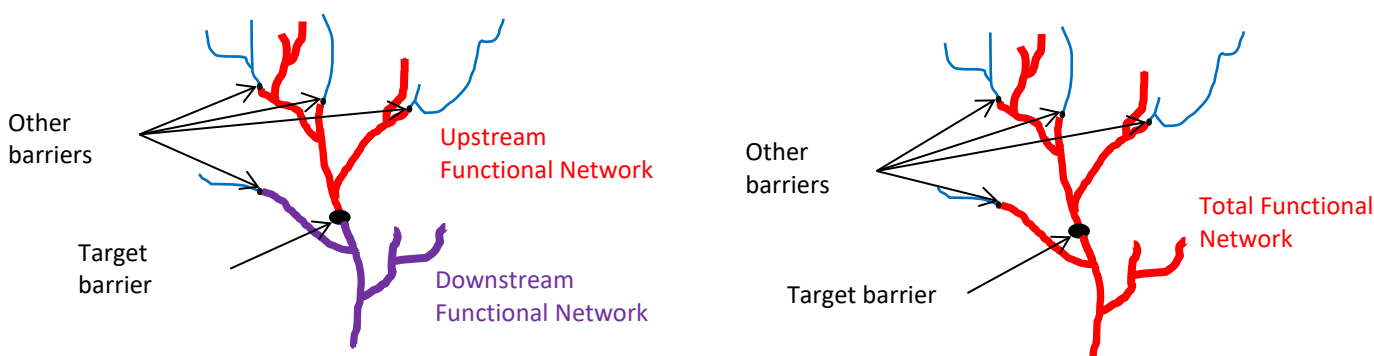


Figure 3-1: Conceptual illustration of functional river networks

3.1.2 Watersheds

For any given barrier, metrics involving three different watersheds are used in the analysis. The contributing watershed, or total upstream watershed, is defined by the total upstream drainage area above the target barrier. Several metrics are also calculated within the local watershed of a target barrier's upstream and downstream functional river networks. These local watersheds are bounded by the watersheds for the next upstream and downstream functional river networks (see Figure 3-2).

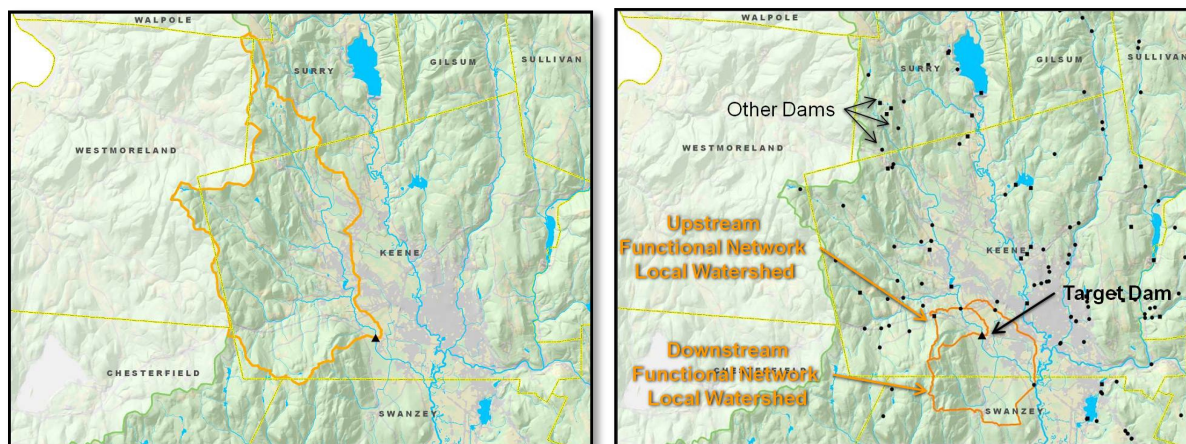


Figure 3-2: Definition of Contributing Watershed. The contributing watershed is defined by the total drainage upstream of a target barrier. The upstream and downstream functional river network local watersheds are bounded by the watershed for the next barriers up and down stream.

3.1.3 Stream size class

Stream size is a critical factor for determining aquatic biological assemblages (Olivero and Anderson 2008, Vannote et al. 1980, Mathews 1998). In this analysis, river size classes, based on the upstream drainage area classes developed for the Northeast Aquatic Habitat Classification System (Olivero and Anderson 2008), were calculated for each segment of the project hydrography and in turn assigned to each barrier (see Figure 3-3).

Drainage area data was not immediately available for the many of the Canadian rivers, nor was it necessary for the analysis which used the Canadian rivers only in so far as they were needed to generate network-related metrics for barriers within Maine.

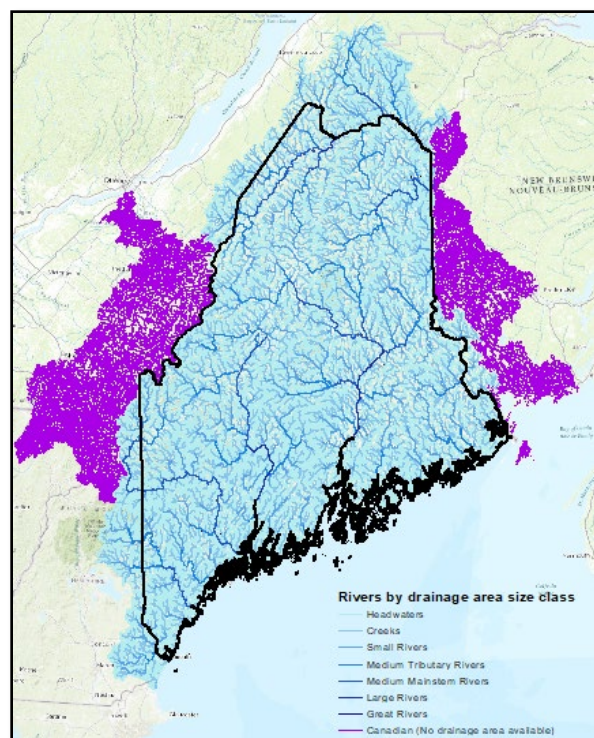


Figure 3-3: Upstream drainage area size class definitions and map of rivers by size class in the Maine.

- 1a) Headwaters (<3.861 mi²)
- 1b) Creeks (>= 3.861<38.61 mi²)
- 2) Small River (>=38.61<200 mi²)
- 3a) Medium Tributary Rivers (>=200<1000 mi²)
- 3b) Medium Mainstem Rivers (>=1000<3861 mi²)
- 4) Large Rivers (>=3861 < 9653 mi²)
- 5) Great Rivers (>=9653 mi²)
- (Defining measure = upstream drainage area)

3.2 Hydrography

In order for barriers to be included in the analysis, they had to fall on the mapped river network, or hydrography, that was used in the project: a modified version of the High Resolution National Hydrography Dataset Plus (NHDPlus HR). This hydrography was digitized by the United States Geological Survey primarily from 1:24,000 scale topographic maps.

In order to be used in this analysis, the hydrography had to be processed to create a dendritic network, or dendrite: a single-flowline network with no braids or other downstream bifurcation (see Figure 3-4). To extract a dendritic network, the following selection query was applied in ArcGIS 10.3:

NHDFlowline.FLOWDIR = 'With Digitized' AND NHDPlusFlowlineVAA.Divergence in (0,1) AND NHDPlusFlowlineVAA.StreamCalc <>0

Further manual modifications of this network were made by Alex Abbott, contractor to the USFWS Gulf of Maine Office and Hadley Couraud, TNC. A dendritic network was also necessary for the portions of Canada that lie downstream of Maine barriers (e.g. Wolastoq-St. John River). Unfortunately, although the High Resolution NHD flowlines are available for Canada, the “Plus” attributes that are required to use the above selection query were not available. Thus, to derive a dendritic network in Canada, a Geometric Network was created from the hydrography in ArcGIS 10.3 so that offending loops and bifurcations could be identified. Each offending section was then manually edited by selecting the mainstem or otherwise removing line segments to create a dendritic network.

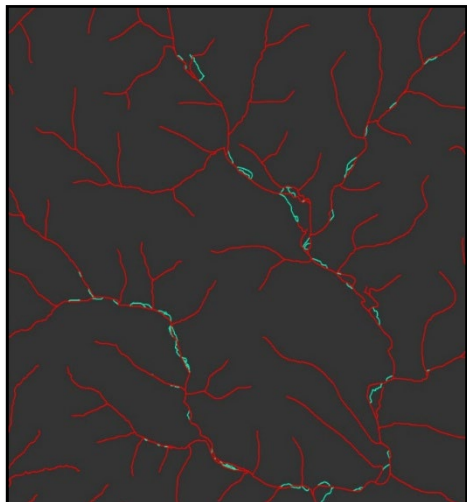


Figure 3-4: Braided segments highlighted in blue needing to be removed to generate a dendritic network.

The result of this process was a single-flowline dendrite, based on the current High Resolution NHD (with some manual modifications), for Maine and downstream portions of Canada. This dendrite (hereafter referred to as the “project hydrography”) was then further processed using custom Python scripts to build a ‘FromNode’ and ‘ToNode’ for each segment in order to establish flow direction for the network. Additional processing using ArcGIS Spatial Analyst and custom Python scripts in ArcGIS was performed to accumulate upstream attributes. This processing produced values including the total upstream drainage area, percent impervious surface, and similar land cover metrics for each line segment.

3.3 Barriers

In-stream barriers used in the analysis included dams, road-stream crossings, and natural barriers, are detailed in the sections that follow.

3.3.1 Dams

The dam data used in the project is maintained by TNC in Maine and was originally sourced from the Maine GIS (MEGIS) (see Appendix III:). In order to perform network analyses in a GIS, the points representing dams and road-stream crossings must be topologically coincident with lines that represent rivers.



Figure 3-5: Illustration of snapping a dam to the river network

Over the course of several years, staff contracted to the USFWS Gulf of Maine Program along with TNC in Maine staff have invested substantial time in reviewing and refining the location of dams to ensure that they were both valid and in the correct location. Various datasets, including a Maine DEP dam dataset and American Rivers dam removal database were used to update this dataset in 2024. This process included “snapping” the dams in a GIS to the project hydrography (see Figure 3-5). For this project, dams were further snapped using ArcGIS tools accessed via Python (arcpy) to ensure they were coincident to the project hydrography. A snapping tolerance of 100m was used, though in the vast majority of cases the dams were already coincident with the river network in which case the dams did not move.

3.3.2 Road-stream crossings

Road-stream crossings can also act as a barrier to fish and other aquatic organisms attempting to move up or downstream. Culverts in particular can, be completely impassible to aquatic organism passage (AOP) due to outlet perch, physical blockages, or elevated flow velocity. The degree to which a road-stream crossing is a barrier to AOP can vary substantially given the physical parameters of the structure and stream conditions. Thus, in order to understand the extent to which a given road-stream crossing is a barrier, it is necessary to survey the crossing from the perspective of AOP.

From 2007 to 2019, The Nature Conservancy partnered with the USFWS, NRCS, Maine Audubon, Maine Forest Service and dozens of volunteer groups to survey all the accessible road-stream crossings around the state of Maine. Based on TNC's review of these data, there were at least 225 individual surveyors who surveyed 969 dams and 25,054 road-stream crossings.

The table below lists the numbers of surveyed dams and road-stream crossings in each barrier severity category. These numbers include tidal crossings, which were all assessed using the standard crossing survey protocols, which included instructions to survey at low-tide to show the most limiting conditions for passage. Though one is in development, a tidal crossing survey protocol was not available at the time of active data collection.

Barrier Status	Number and Percent of Crossings and Dams
Definite Barrier	7,994 (29%)
Potential Barriers (at least half of which are 'definite barriers')	7,945 (29%)
No Barrier	8,257 (30%)
Unknown	3,340 (12%)

Road Class	Number and Percent of Crossings
Town Roads (including State Assist)	5,903 (22%)
State Roads	5,754 (22%)
Federal (Interstate)	424 (2%)
Private Roads*	13,720 (52%)
Railroads	633(2%)

**Private roads are primarily in Maine's large private timberland ownerships.*

3.3.2.1 Private crossings

The Nature Conservancy has signed data sharing agreements with several large landowners in northern Maine. These data sharing agreements allow TNC to survey crossings on these lands and to use the data for planning purposes. The data themselves, however, cannot be shared or displayed publicly.

These restricted road-stream crossing data points are used in the process of calculating metrics but are not included in the results nor made available for public viewing in the tool. For example, if a public

crossing has two other public crossings and three restricted private crossings downstream of it, then the metric “number of downstream barriers” would equal five. When the results are viewed in the map, however, only the two public crossing points would be visible. Similarly, a barrier’s upstream functional network (see Section 3.1.1) and any derived metrics are calculated based on the presence of restricted barriers.

3.3.2.2 Canadian barriers

The Canadian barriers used in our analyses were obtained from Nature Conservancy Canada from their Aquatic Connectivity Tool for the Northern Appalachian–Acadian Region of Canada (Noseworthy et al 2019). This dataset, which included dams as well as road-stream crossings, was treated in much the same way as restricted barriers. These barriers were used in the generation of metrics to understand their impact on the public crossings in Maine, but they are not included in the results.

3.3.2.3 Road Classes

When referencing road class in the report and tool, ‘public’ includes any and all town, county, state, and federal roads – those managed and maintained by those respective governmental entities. ‘Private’ roads include roads owned and maintained by private individuals and entities. Importantly, in the landscape of Maine, the majority of road miles identified as ‘private’ belong to Maine’s large private timberland ownerships.

3.3.3 Natural Barriers

Natural barriers, such as waterfalls, can also act as barriers to fish passage. Including them in the analysis was important due to the impact barriers have across a network. For example, a waterfall just upstream of a barrier could drastically affect the length of that barrier’s upstream functional network, or the amount of habitat that would be opened by removing that barrier. Thus, although waterfalls are excluded from the project results (they are not intended to be removed or bypassed) they were included in the generation of functional networks.

The primary data source for waterfalls was the Maine stream barrier database. This database includes natural barriers that have been surveyed by field crews as part of the road-stream crossing survey efforts and classified as barriers, potential barriers, or non-barriers. Barriers and potential barriers were used in the generation of the functional river networks.

3.3.4 Barrier Passability

All barriers were assigned a “passability” score ranging from 0 – 1 to describe the degree to which that barrier is passable to aquatic organisms. Barriers classified with a score of 0 were considered complete barriers while a score of 1 were considered fully passable and omitted from the prioritization analysis. A bridge or a fully passable culvert with natural substrate are examples of crossings with a passability score of 1. As no further upgrades could be made to these structures, there is no reason to prioritize them for upgrade, nor to assess their influence on other barriers. Scores in between 0 and 1 describe varying degrees of partial passability. Barriers with a passability score less than 1 were used to generate functional river networks for the analysis. Dam and road-stream crossing passability scores were

calculated separately, based on different parameters and methods. The next two sections detail the methods and parameters used for each.

3.3.4.1 Dam passability

There is currently no authoritative scoring dataset or methodology for assigning passability scores at dams. Given this lack of data, passability at dams was estimated based on known fish passage facilities and a simple rubric to score them on a 0 - 1 scale.

- **0** – No passage facility, complete barrier. (Default assumption if no other info)
- **0.25** – Only the best swimmers can do it, some of the time. We know there's a facility, but we know little about its effectiveness. (e.g. Bolster's Mill Dam on the Crooked River)
- **0.5** – Average passage facility. Facility works okay, not the best; temporal passage, moderate amounts, for several species but likely not all (e.g. Brunswick Dam)
- **0.75** – State of the art, known to be effective, maybe not at all times. (e.g. Milford lift)
- **1.0** – Totally passable. (e.g. Howland bypass)

Dam passability scores were originally assigned (for the 2018 tool) by TNC staff based on a review of dams in the state and were further made available for review by partners at MDIFW Region A via web map. For the 2025 update of the tool, TNC conducted an extensive update of the dams dataset, cross-referencing the data used in the 2018 tool with data from Maine DEP and American Rivers. For the most part, these updates involved an assessment of dams that had been removed or were missing from the dataset but present in other (especially the Maine DEP) datasets. In some cases, previously assigned passability scores were changed based on these edits and, in a few cases passability scores were changed based on information from Maine DMR. Fish passage facility effectiveness was incorporated into the dataset as possible. The assessment was done entirely remotely, aimed at capturing full dam removals; no field verification was conducted.

3.3.4.2 Road-stream crossing passability

Road-stream crossings were first classified into three categories based on the physical condition of the crossing as described in Figure 3-6. Crossings with significant outlet drops (waterfalls) or debris blockages are assumed to have no aquatic organism passage (AOP; passability = 0), crossings with no observable impediments to passage that are greater than or equal to the bankfull width of the stream were assumed to be fully passable (passability = 1). All other crossings were catalogued as having reduced passage due to survey indicators of higher velocities in the crossing than observed in the stream channel (perched inlet/outlet, lack of substrate in the crossing, etc).

In the data and in the tool, the associated Barrier Class terms are:

- 'Full AOP' is called 'No Barrier'
- 'Reduced AOP' is called 'Potential Barrier'
- 'No AOP' is called 'Barrier'
- If the barrier status could not be calculated, the barrier class is 'Unknown'

Metric	Flow Condition	Crossing Classification		
		Full AOP	Reduced AOP	No AOP
		<i>If all are true</i>	<i>If any are true</i>	<i>If any are true</i>
Inlet Grade		At Stream Grade	Inlet Drop or Perched	
Outlet Grade		At Stream Grade		Cascade, Free Fall onto Cascade
Outlet Drop to Water Surface		= 0		≥ 1 ft
Outlet Drop to Water Surface/ Outlet Drop to Stream Bottom				> 0.5
Inlet or Outlet Water Depth	Typical-Low	> 0.3 ft		< 0.3 ft w/Outlet Drop to Water Surface > 0
	Moderate	> 0.4 ft		< 0.4 ft w/Outlet Drop to Water Surface > 0
Structure Substrate Matches Stream		Comparable or Contrasting		
Structure Substrate Coverage		100%	< 100%	
Physical Barrier Severity		None	Minor or Moderate	Severe

Figure 3-6: AOP at road-stream crossings classification system (Abbott. 2016. NAACC Stream Crossing Survey Data Form Instruction Guide Version 1.2 Guidance. North Atlantic Aquatic Connectivity Collaborative. University of Massachusetts Amherst.)

When applying this classification system to the surveyed data of more than 25,000 crossing point across the state of Maine, we observed that more than half of the crossings in the state fall into the reduced AOP /velocity barrier classification, requiring a deeper level of analysis to better identify the severity of these velocity barriers and improve our understanding of which crossings would best benefit fish passage by replacement with rightly sized structures.

Refinement of Reduced AOP Classification:

We re-analyzed the reduced AOP subset of “potential barrier” crossings to derive a rough quantitative estimate of the average velocity in the crossing for at given flow. Streamstats (USGS 2016) outputs for bankfull (peak 1.1 Year recurrence interval) and low flow (Mean 10 year 7 day) flows were used to bound the upper and lower limit and define a range of flows that fish could experience when migrating.

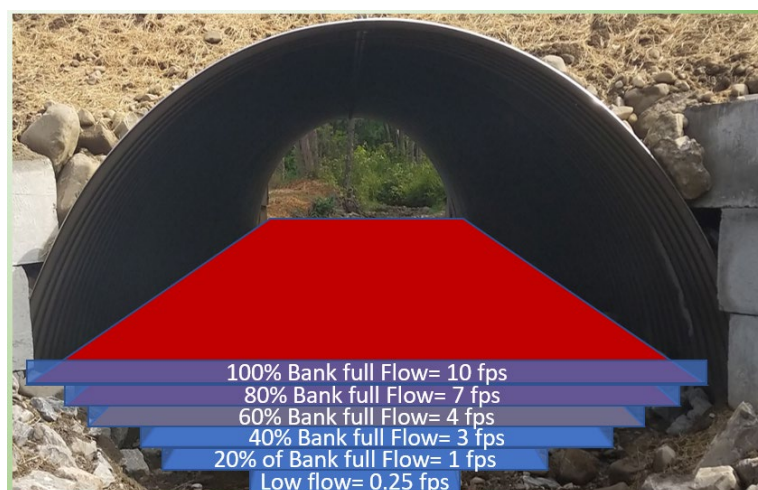


Figure 3-7: Diagram of flow percentile stratification and example velocity outputs.

This flow range was broken into 10% intervals and input into Manning's equation to develop an estimate of average flow velocity through the crossing at each interval. A threshold of 2 feet per second was chosen as a low enough velocity that we could be confident that most fish species and life stages could navigate without significant metabolic impacts. When this threshold velocity was exceeded, the corresponding flow percentile was reported as the crossings passability score, based on the assumption that aquatic organism passage efficacy would increase as velocities stay below the threshold at bankfull or near bankfull flows.

Caution! This analysis contains compounded error vectors from the StreamStats Inputs, remote sensing of reach slope from NHD and qualitative field assessments methods. This method should not be used for anything other than a course screening method to identify the highest priority sites, and these errors should be assessed carefully when drawing conclusions from the model outputs.

3.3.4.3 Natural barrier passability

Natural barriers were simply assigned passability scores based on their general classification: barriers were assigned a 0 while potential barriers were assigned a score of 0.5. Natural barriers classified as "No Barrier" were assigned a 1 and excluded from the analysis.

4 Analysis Methods

The conceptual framework of the Maine ABPT project rests on a suite of ecologically relevant metrics calculated for each barrier in the analysis. These calculated metrics are then used to rank the relative benefit of removing or improving passage at any given barrier point in the network. At its simplest, a single metric could be used to evaluate barriers. For example, if one is interested in passage projects to benefit diadromous fish, then the barrier's upstream functional network, or the number of river miles that would be opened by that barrier's removal, could be used to prioritize dams. In this case, the barrier with the longest upstream functional network—the barrier whose removal would open up the most river miles—would rank at the top of the list. Metrics can be combined and weighted to indicate the relative importance of each metric in a set of species-based scenarios, as described in further detail in Section 4.3.

4.1 Geographic Stratification of Analyses

Each scenario-based prioritization (described in subsequent sections) was run at a statewide extent, using the applicable input barriers for each scenario (e.g. see Section 4.3.1). Each scenario-based prioritization was also run separately for each HUC 8 and HUC 10 watershed in the state. This allows users to view prioritization results relative to the entire state or relative to their watershed of interest. For example, if a user wanted to understand how barriers were prioritized for inland Eastern Brook trout, they could view the results of the Inland Brook Trout scenario at a statewide scale, a large watershed scale (e.g. the Androscoggin HUC8), or a smaller watershed scale (e.g. the Little Androscoggin HUC10) – each of these scales is their own scenario-based prioritization.

4.2 Metric Calculation

A total of 82 metrics were calculated for each barrier in the study area using ArcGIS methods. The process used to generate each metric was scripted using Python 3.9 primarily using the arcpy 3.2 module (ArcGIS Python package) while incorporating other Python packages as needed.

Metrics are organized into six categories for convenience: Network, Invasive Species, Landscape, inland Eastern Brook trout, Atlantic salmon, and Coastal Anadromous. These categories help organize the metrics into a logical order *but have no impact on the analysis*. Additionally, in the analyses, each metric is sorted in either ascending order or descending order to indicate whether large values or small values are desirable in a given scenario. For example, ‘upstream functional network length’ is sorted descending because large values are desirable – a passage project on a barrier that opens up more river miles is desired over a passage project which opens up few miles. Conversely, ‘percent impervious surface’ is sorted ascending because small values are desirable – a passage project that opens up a watershed that has little or no impervious surface is desired over a project that opens up a watershed with a high percentage of impervious surface. A table listing each of the metrics is presented in Figure 4-1, and a more complete description of each metric can be found in Glossary of Metric Definitions (Appendix IV).

Depending on the objectives of a prioritization scenario, some metrics will be of greater importance than other metrics. The upstream functional network length may be of particular interest in a prioritization scenario focused on diadromous fish, for example, while the percent impervious surface in the floodplain of the upstream functional river network may be of less importance, and the presence of a rare crayfish species may be of no interest. Relative weights, which must sum to 100, were assigned to each metric to indicate its importance in a given scenario. Figures 4-2 through 4-6 list the five scenarios identified by the original Workgroup.

Metric weights are subjective in nature; there are no hard and fast rules regarding how to properly select and weight metrics for a given target species like alewife. To arrive at the weights presented in the tables below, the Workgroup went through an iterative process of selecting draft weights based on their knowledge of the species of interest, then adjusting them in light of draft results produced from the selected weights and their current on-the-ground removal priorities. This process was intended to help calibrate the results to known priorities.

Category	Metric Name	Units	Sort Order
Network	Upstream Functional Network Length	m	D
	Count of Downstream Barriers	#	A
	Absolute Gain (min of upstream & downstream functional network)	m	D
	Total length of upstream river network (Ignoring barriers)	m	D
	Total Functional Network Length (upstream + downstream)	m	D
	Density of barriers on upstream network	#/m	D
	Count of downstream natural barriers	#	A
	Count of downstream fish passage facilities	#	D

Invasive Species	Count of downstream hydropower facilities	#	A
	River Size Class (based on drainage area)	unitless score	A
	One of upstream barriers is a natural barrier	Boolean	A
	The product (multiplication) of downstream barrier passability scores	unitless score	D
	Invasive species are found in the network on one side of a barrier and not the other	Boolean	A
	Invasive species (confirmed occurrences) are found in the network on one side of a barrier and not the other	Boolean	A
Landscape	Percent Forested Landcover in upstream drainage area	%	D
	% forested landcover in 100m buffer of upstream network	%	D
	% forested landcover in 100m buffer of downstream network	%	D
	% forested landcover in 100m buffer of total network (upstream & downstream)	%	D
	% medium/high intensity developed landcover in 100m buffer of upstream network	%	A
	% medium/high intensity developed landcover in 100m buffer of downstream network	%	A
	% medium/high intensity developed landcover in 100m buffer of total network (upstream & downstream)	%	A
	% cultivated crops in 100m buffer of upstream network	%	A
	% cultivated crops in 100m buffer of downstream network	%	A
	% cultivated crops in 100m buffer of total network (upstream & downstream)	%	A
	% pasture/hay in 100m buffer of upstream network	%	A
	% pasture/hay in 100m buffer of downstream network	%	A
	% pasture/hay in 100m buffer of total network (upstream & downstream)	%	A
	% forested landcover in floodplain of upstream network	%	D
	% forested landcover in floodplain of downstream network	%	D
	% forested landcover in floodplain of total network (upstream & downstream)	%	D
	% medium/high intensity developed landcover in floodplain of upstream network	%	A
	% medium/high intensity developed landcover in floodplain of downstream network	%	A
	% medium/high intensity developed landcover in floodplain of total network (upstream & downstream)	%	A
	% cultivated crops in floodplain of upstream network	%	A
	% cultivated crops in floodplain of downstream network	%	A
	% cultivated crops in floodplain of total network (upstream & downstream)	%	A
	% pasture/hay in floodplain of upstream network	%	A
	% pasture/hay in floodplain of downstream network	%	A
	% pasture/hay in floodplain of total network (upstream & downstream)	%	A
	Barrier is on conservation land	Boolean	D
	% Conserved land in 100m buffer of upstream functional network	%	D
	% Conserved land in 100m buffer of upstream functional network	%	D

	Aquifers are found in upstream functional network OR downstream functional network OR both (all treated the same)	Boolean	D
	Aquifer OR coarse sediments are found in upstream functional network OR downstream functional network OR both (all treated the same)	Boolean	D
	Calcareous or moderately calcareous geology is found in upstream functional network OR downstream functional network OR both	Boolean	D
Brook trout	Probability of BKT occupancy (current temps)	%	D
	Probability of BKT occupancy (+2C)	%	D
	Probability of BKT occupancy (+4C)	%	D
	Probability of BKT occupancy (current temps) >= 50%	Boolean	D
	Probability of BKT occupancy (+2C) >= 50%	Boolean	D
	Probability of BKT occupancy (+4C) >= 50%	Boolean	D
	Miles of high or very high priority Brook trout habitat in upstream + downstream networks	miles	D
	Miles of high or very high priority Brook trout habitat in upstream functional network	miles	D
	Presence of high or very high Brook trout habitat upstream OR downstream of barrier	Boolean	D
	Presence of >=1/4 mile very high priority Brook trout habitat upstream OR downstream of barrier	Boolean	D
	Barrier is within EBTJV Wild Brook trout Patches	Boolean	D
	Heritage fish ponds in upstream functional network AND downstream functional network	Boolean	D
Atlantic salmon	Cumulative upstream salmon habitat units, discounted by barrier pass	unitless score	D
	Cumulative upstream network length, discounted by barrier pass	unitless score	D
	Salmon Habitat Quality streamline that the barrier is on	unitless score	D
	Barrier is in a Salmon Critical Habitat HUC10	Boolean	D
	Summed modeled salmon habitat units in the HUC10 the barrier is within	unitless score	D
	Number salmon habitat units in US Functional network	unitless score	D
	Sum of salmon parr productivity in US functional network	unitless score	D
	Upstream salmon restoration potential	unitless score	D
Coastal Anadromous	Miles of tidal habitat (current) upstream of barrier	miles	D
	Miles of tidal habitat (HAT +3.9ft) upstream of barrier	miles	D
	Miles of tidal habitat (HAT +6ft) upstream of barrier	miles	D
	Barrier falls on sea-run Eastern Brook trout lines	Boolean	D
	Barrier is located on a tidal stream (HAT +3.9ft)	Boolean	D
	Upstream acres of alewife ponds (total, ignoring any barriers in the way)	acres	D
	There are smelt sites upstream of the barrier	Boolean	D

Figure 4-1: Metrics calculated for each barrier in the study

4.3 Fish Prioritization Scenarios

The metrics noted in Figure 4-1 serve as a menu of metrics by which barriers can be assessed. These metrics were weighted and combined (see Section 4.4 for details of prioritization) to produce five different fish prioritization scenarios that were identified by the original project workgroup (see Appendix II). In some cases, scenarios only included a subset of barriers in the state. For example, the Coastal Anadromous Fish scenario only included those barriers on coastal streams. In other cases, all barriers in the state were included.

Each fish prioritization scenario was run for the entire set of applicable barriers throughout the state and as subsets stratified by HUC8 and HUC10 watersheds. This allows users to see scenario priorities displayed relative to the state or to their watershed of interest.

The following tables list the fish prioritization scenarios - the weights that were given to metrics in each scenario and any filter that was applied to barriers to limit which barriers were included.

4.3.1 Coastal Anadromous Fish

The Coastal Anadromous Fish scenario was designed to prioritize barriers whose removal could benefit a suite of coastal anadromous fish species including sea-run Eastern Brook trout and Rainbow smelt (*Osmerus mordax*). The analysis was limited to barriers falling on small coastal streams. Coastal streams were defined as streams with an upstream drainage area less than 100 km² that drain into a tidal area (directly into the ocean or the tidal portion of a larger river).

Metric Name	Coastal Anadromous
Upstream Functional Network Length	10
The product (multiplication) of downstream barrier passability scores	10
Percent Forested Landcover in upstream drainage area	10
% forested landcover in 100m buffer of total network (upstream & downstream)	10
Aquifer OR coarse sediments are found in upstream functional network OR downstream functional network OR both (all treated the same)	15
Calcareous or moderately calcareous geology is found in upstream functional network OR downstream functional network OR both	5
Barrier falls on sea-run Brook trout lines	5
There are smelt sites upstream of the barrier	10
Miles of high or very high priority Brook trout habitat in upstream + downstream networks	15
Miles of high or very high priority Brook trout habitat in upstream functional network	10

Figure 4-2: Metric weights for the coastal anadromous fish scenario

4.3.2 Inland Brook trout

The Inland Brook trout scenario was designed to assess barriers whose removal or improvement could benefit inland Eastern Brook trout. It was designed to provide a counter to the coastal anadromous fish scenario and so explicitly excludes the coastal streams that were identified for the coastal anadromous scenario.

Metric Name	Inland Brook trout
Upstream Functional Network Length	10
Absolute Gain (min of upstream & downstream functional network)	10
Total Functional Network Length (upstream + downstream)	5
Aquifer OR coarse sediments are found in upstream functional network OR downstream functional network OR both (all treated the same)	10
Calcareous or moderately calcareous geology is found in upstream functional network OR downstream functional network OR both	5
Percent Forested Landcover in upstream drainage area	10
% forested landcover in 100m buffer of total network (upstream & downstream)	10
Probability of BKT occupancy (+2C)	10
Probability of BKT occupancy (+2C) >= 50%	5
Miles of high or very high priority Brook trout habitat in upstream + downstream networks	10
Miles of high or very high priority Brook trout habitat in upstream functional network	5
Presence of high or very high Brook trout habitat upstream OR downstream of barrier	5
Heritage fish ponds in upstream functional network AND downstream functional network	5

Figure 4-3: Metric weights for the Inland Brook trout scenario

4.3.3 Atlantic salmon

The Atlantic salmon scenario is focused on identifying barriers to benefit the Endangered Species Act-listed Atlantic salmon. To be included in the salmon scenario, crossings had to meet the following criteria:

- If a crossing fell within the Distinct Population Segment (DPS), they were pre-filtered using the Critical Habitat HUC10s. Only crossings in one of these HUC10s were scored and prioritized.
- If a crossing fell outside the DPS, they were pre-filtered using the HUC10s that contain streamlines identified as Medium, High, or Very High quality Atlantic salmon habitat. Only crossings in one of these HUC10s were scored and prioritized.

Barriers that met these criteria were prioritized using the following metrics and weights.

Metric Name	Atlantic salmon
Upstream Functional Network Length	20
Count of Downstream Barriers	10
Count of downstream fish passage facilities	10
The product (multiplication) of downstream barrier passability scores	10
Percent Forested Landcover in upstream drainage area	10
Cumulative upstream salmon habitat units, discounted by barrier pass	15
Salmon Habitat Quality streamline* that the barrier is on	15
Barrier is in a Salmon Critical Habitat HUC10 watershed	10

*Some areas depicted on the Salmon Habitat Quality streamline layer are outside of the Distinct Population Segment and State/Federal active management areas for the endangered population of Atlantic salmon.

Figure 4-4: Atlantic salmon scenario metrics

4.3.4 Alewife

Given the importance of alewife spawning ponds to alewife in Maine, the alewife scenario was run for those barriers downstream of mapped alewife spawning ponds, whose removal could improve access to these critical ponds.

Metric Name	Alewife
Count of Downstream Barriers	10
Count of downstream fish passage facilities	20
The product (multiplication) of downstream barrier passability scores	10
Percent Forested Landcover in upstream drainage area	20
Upstream acres of alewife ponds (total, ignoring any barriers in the way)	40

Figure 4-5: Alewife scenario metrics

4.3.5 American shad / Blueback herring

Finally, American shad and Blueback herring were assessed via a separate scenario from alewife using all barriers in the state.

Metric Name	American shad & Blueback herring
Upstream Functional Network Length	20
Count of Downstream Barriers	10
Count of downstream fish passage facilities	20
Count of downstream hydropower facilities	20
The product (multiplication) of downstream barrier passability scores	10
Percent Forested Landcover in upstream drainage area	20

Figure 4-6: American shad / Blueback herring metrics

4.4 Prioritization

Once metric values were calculated and relative weights assigned to the metrics of interest for each prioritization scenario, metrics were combined through a weighted ranking process to develop a prioritized list for each scenario for each geography (statewide, HUC8, HUC10).

This process can be expressed as:

For metric value a in set $A = \{a_1, a_2, a_3, a_n, \dots\}$

R_a = ordinal number of value a in set A , sorted ascending or descending, depending on metric. Dense ranking, where tie values are both given the same rank and the following value is given the next sequential rank (e.g. '1223') was used for ties.

$$PR_a = 100 \left(\frac{R_a - \min A}{\max A - \min A} \right) \quad (\text{descending})$$

or

$$PR_a = 100 - \left(100 \left(\frac{R_a - \min A}{\max A - \min A} \right) \right) \quad (\text{ascending})$$

$$CS = \sum_{i=1}^m W_i PR_i$$

$$FR = R_{CS}$$

Where:

m = metric

R = rank

CS = Combined Score

W = weight

PR = percent rank

FR = final rank

4.4.1 Developing ranking for each prioritization scenario and geography

1. All metric values were normalized to a percent scale where the optimal value is assigned a score of 100 and the least desirable value is assigned a score of 0
2. All metric values for a prioritization scenario were multiplied by their assigned weight in the given scenario (e.g. 15%) to produce a weighted rank for each metric
3. The weighted ranks were then summed to create a summed rank
4. The summed ranks were then ranked to create a final ranked result
5. The final ranked results were then assigned to tiers/bins for display in the web map application.

4.4.2 Binning rules for final ranked results

If there were **at least 10 barrier points** in the set being evaluated for a given prioritization scenario in a given geography, then the final ranked results were binned into 5 tiers representing the following percentages of the data.

- Tier 1 captures the top 10% of the results, the highest priority
- Tier 2 captures the next 10-25% of the results
- Tier 3 captures the next 25-50% of the results
- Tier 4 captures the next 50-75% of the results
- Tier 5 captures the final 75-100% of the results

Most prioritization scenario / geography combinations had at least 10 points but in some smaller geographies there were fewer than 10 points. If there were fewer than 10 barrier points in the set being evaluated for a given prioritization scenario in a particular geography, then the ranked results were binned into 2 tiers based on the midpoint of the ranks. If there 2 or fewer points in the set, the bin number was simply directly assigned from their rank sum (1 or 2).

This was done because although the prioritization produces a sequential list of prioritized barriers, the precision with which metrics can be calculated in a GIS is not necessarily indicative of significant ecological differences. The workgroup was keen to ensure that the ranking not portray the results with more ecologically significant nuance than is appropriate.

4.5 Additional Features and Analyses

4.5.1 Upstream and Downstream River Network Tracing

When the user clicks on a Barrier on the web map, along with all the detailed information that is presented, the network flowlines upstream and downstream of the barrier are traced and highlighted in red and orange. These up and downstream trace lines continue all the way up and down the network (to the top of the watershed and to the ocean), regardless of intervening barriers. This feature is designed to show the subject barrier's place and role in its overall network / watershed. ArcGIS Pro / arcpy trace network tools were used to conduct this analysis with a trace-network enabled version of the Hydrography (2023) (the same hydrography dataset used for all network metric calculations.)

4.5.2 Flood Risk

Important Note: This section was written in and describes the flood risk analysis originally run in 2020. Neither this text nor the analysis were updated for the 2026 version of the prioritization tool.

There is a correlation between culverts that are undersized for floods and problems for fish passage. O'Shaughnessy et al (2016) suggests that there are dual benefits to replacing undersized culverts in Maine that are both barriers to fish passage and unable to pass a 25-year flood event.

Flood risk at culverts was estimated using the conceptual process outlined in Figure 4-7. Using flow data from StreamStats (USGS 2016), culvert survey data from Maine's stream crossing surveys, and LiDAR derived elevations and a river hydrography, the likelihood of a given road being overtopped within 30 years was calculated as described below. Not all crossings in the database could be run through the flood risk analysis, typically due to missing key information necessary for the calculations.

The results found that 4,580 culverts out of the surveyed stream crossings in Maine are barriers to fish passage and likely overtop during a 25-year event. The efficacy of a given road-stream crossing to allow the passage of fish and other aquatic organisms was determined using a barrier screening model that gives an AOP score based on various input variables. While many culverts in the state need to be addressed, focusing on those that present a flood risk to humans and a barrier to fish species enables road owners to prioritize crossings that have multiple benefits and provides other funding sources.

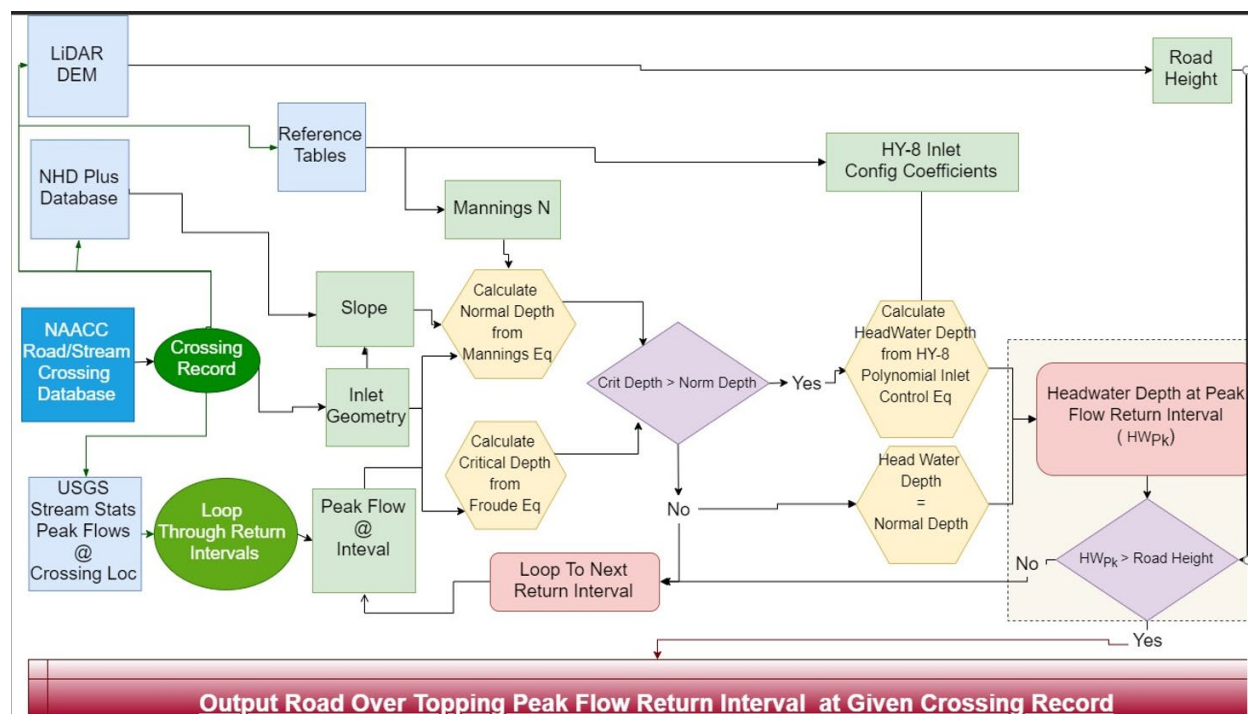
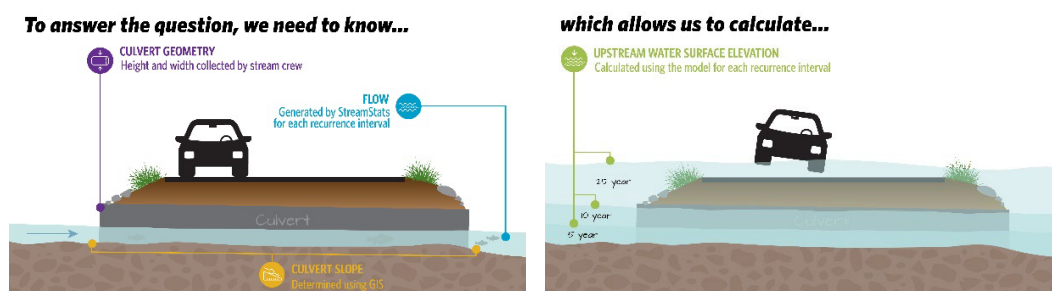


Figure 4-7: Flowchart of Flood Risk Programming Method

Using the measured crossing geometry from the North Atlantic Aquatic Connectivity Collaborative (<https://naacc.org/>) and querying USGS StreamStats website (<https://streamstats.usgs.gov/ss/>) for Flood Flows estimates of 1, 2, 5, 10, 25, 50, 100, 250 and 500 year events, we are able to solve polynomial equations derived for Federal Highways Administration HY-8 Model to solve for the headwater depth for each peak flow at a given crossing.



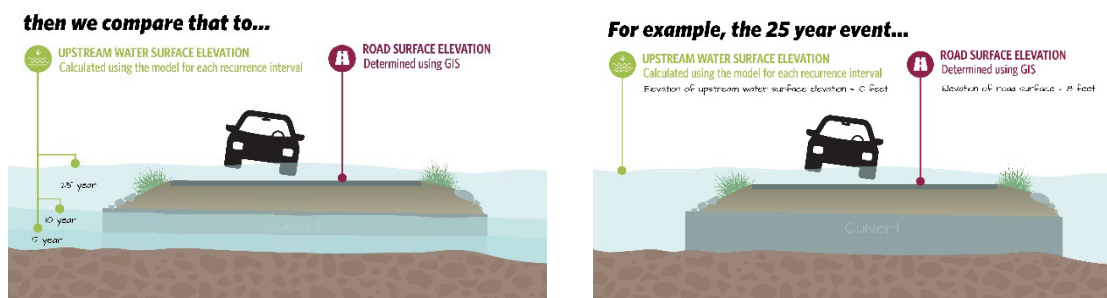


Figure 4-8: Graphic Description of flood risk calculation methods

Comparing the headwater depth to the height of the road above the stream bottom at the crossing inlet (from LiDAR data analysis conducted by TNC's spatial scientist, Dan Coker), we can estimate what peak flow recurrence interval would likely overtop the road and calculate the probability of this overtopping event happening in the next 30 years. Figure 4-8 provides a graphic representation of the overall analysis.

This flood risk analysis results are presented as a complimentary component to the ecological prioritization with the intent that a culvert that is both a high priority for one or more ecological scenarios and is also a high-risk for flooding would be a good candidate project that could address multiple concerns and potentially bring in additional partners and funding (e.g. town roads commissions, Maine DOT).

5 Using the Tool

5.1 Tool Organization

Upon first entering the new Aquatic Barrier Prioritization Tool, users are presented with the Help page open to an introductory Welcome and a disclaimer about the use and limitations of the tool. The Help page, which can be accessed from a button on the header, regardless where on the tool a user is navigating, also provides:

- Tool Tutorial – an abbreviated version of how to use each tab of the tool.
- Comprehensive Tool Report – a description & link to download a PDF of the full tool report (this document).
- Data Sources Appendix - a description & link to download a PDF of the Data Sources Appendix.
- Metrics Glossary - a description & link to download a PDF of the Glossary of Metric Definitions.
- Fish Scenarios – descriptions and tables depicting the metrics and associated weights used in the fish prioritization scenarios.
- FAQ – a selection of Frequently Asked Questions and their respective answers.

Start by selecting a specific geography (either at the Statewide, HUC 8 or HUC10 watershed scale) within which you want to prioritize the physical barriers to fish passage. A user can select the watershed from the dropdown menu, or by selecting the watershed boundary directly on the map. Then, tab through the options in the top bar to select and view prioritizations within that geography for different fish species, to filter results by additional metrics, and finally to see the site-specific data for an individual barrier. The user can view additional political and ecological layers concurrently with the results in the 'Layers' tab.

NEXT: GO TO FISH PRIORITY ▶

Welcome to the Maine Aquatic Barrier Prioritization Tool

This tool is designed to support the river restoration community in identifying in-stream barriers for removal or improved fish passage. The prioritized results are designed to help identify high priority fish passage projects. It is based on the best available state-wide data and does not incorporate any data on social, feasibility, or economic factors. It is intended to provide screening-level support and should not be considered a substitute for local-scale information.

This tool builds on the Penobscot Aquatic Barrier Prioritization tool, which was funded by the NOAA Office of Habitat Conservation through the NOAA Habitat Blueprint, Penobscot Habitat Focus Area (beginning in 2014). This material is based upon work supported by the Natural Resources Conservation Service, U.S. Department of Agriculture, under number NR181218XXXXG001(YYYY). Any opinions, findings, conclusions, or recommendations expressed in this publication are those of the author(s) and do not necessarily reflect the views of the U.S. Department of Agriculture.

There are three additional functions along the page header:

- TNC Sign In
- Help – *see description above*
- Data Export
 - This function allows the user to download the crossing and prioritization results data to a CSV file. The data exported will be only that which is currently filtered or selected by the user. Additionally, the user can select which categories of metrics they want provided for their exported data.
- Search by crossing Site ID
 - Users can type in a unique Site ID to zoom to and pull up that crossing's data.
- Turn On Site ID labels
 - Users can select this option to have Site ID numbers displayed on the screen.

Select Geography

SHOW SITE ID

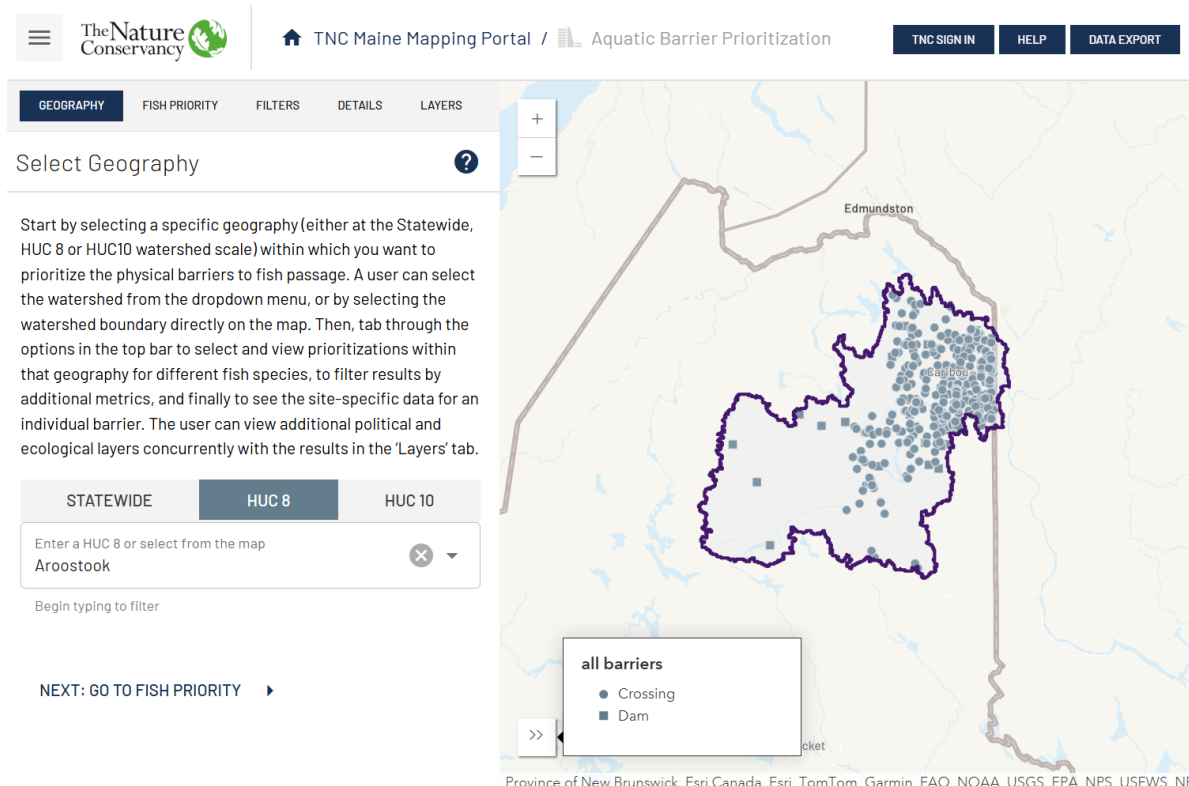
There are 5 tabs within the tool, along the left-hand side of the screen:

- Geography
- Fish Priority
- Filters
- Details
- Layers

5.1.1 Geography Tab

Start by selecting a specific geography (either Statewide, HUC 8, or HUC10 watershed scale) within which the user wants to prioritize the physical barriers to fish passage. A user can select the watershed from the dropdown menu, by typing the name of the watershed in the box, or by selecting the watershed boundary directly on the map.

With a geography selected, users can tab through the options in the top bar to select and view prioritizations within that geography for different fish species, to filter results by additional metrics, and finally to see the site-specific data for an individual barrier. The user can view additional political and ecological layers concurrently with the results in the 'Layers' tab.

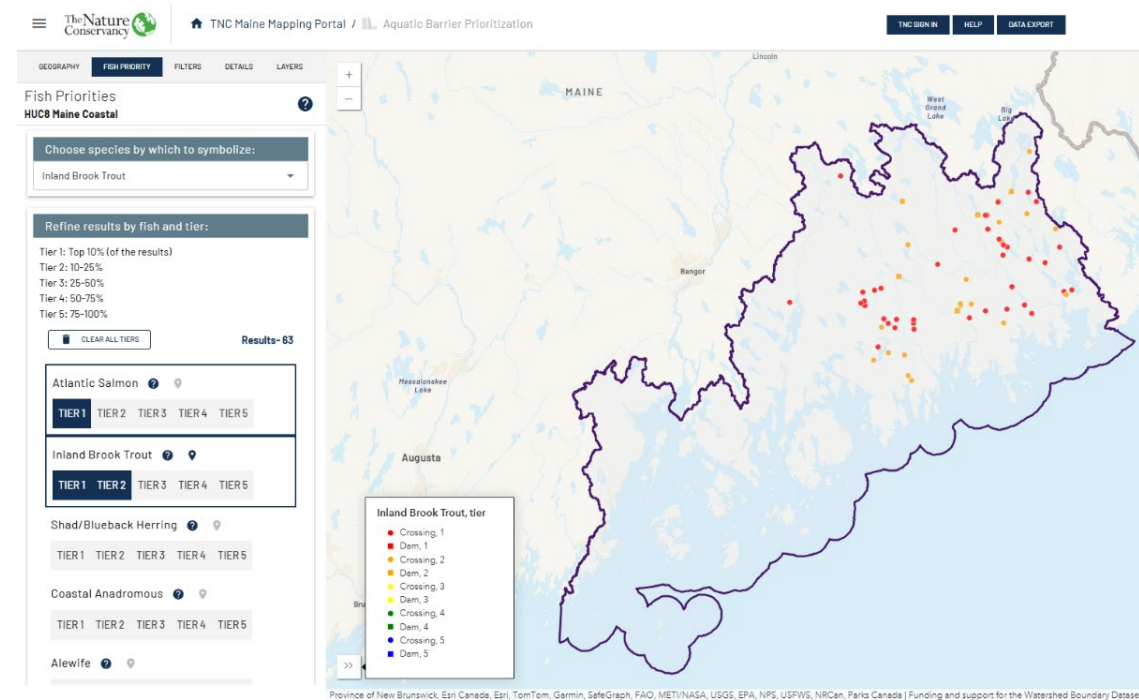
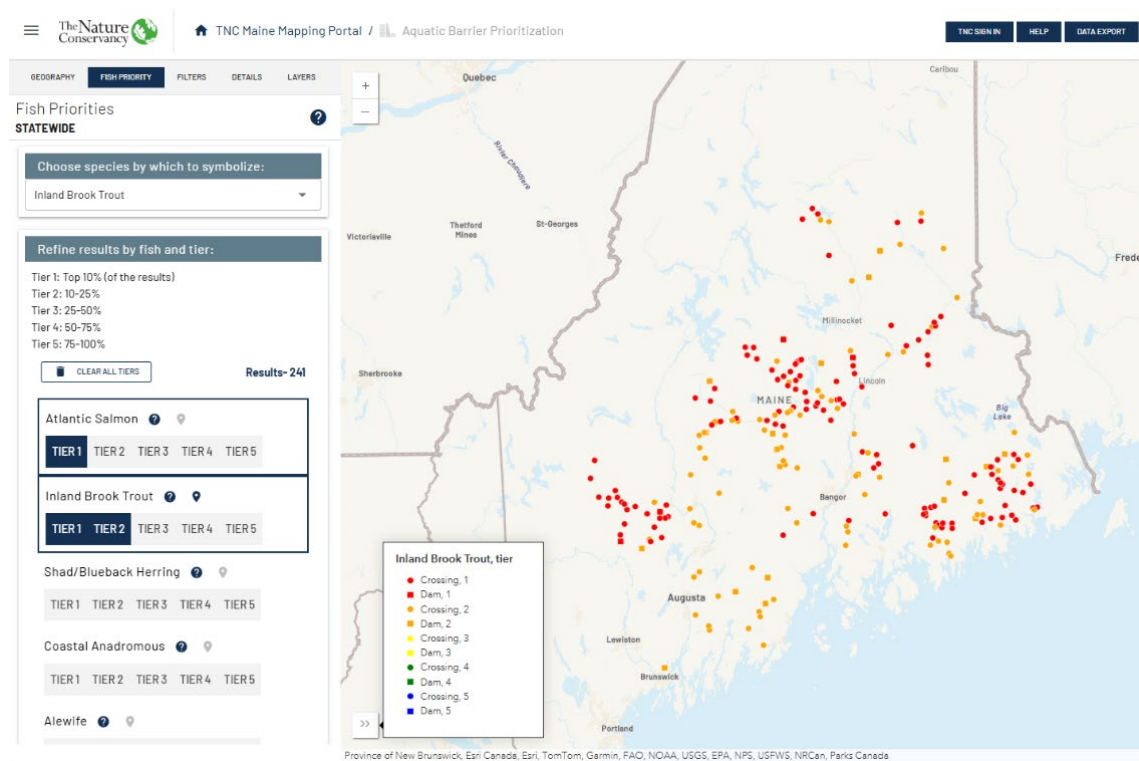


5.1.2 Fish Priority Tab

Select the tiers of prioritized results for the fish species of interest, which can include multiple fish species concurrently. The user can select these tiers by clicking on the respective tier buttons below each fish species and can select multiple at the same time. Results displayed will indicate those barriers that are ranked in each of the species' tiers selected. For example, selecting Tier 1 and 2 for alewife and Tier 1 and 2 for American shad/Blueback herring (ASBH) will only display crossings that are both Tier 1 or 2 for alewife **and** Tier 1 or 2 for ASBH. This means that as the user adds tiers of different scenarios, the criteria for being visible get more restrictive; they are not additive. Crossings have to satisfy all the tiers the user has selected, not just be in one of them.

The species chosen for symbolization will show how the crossings in those tiers rank for that species. (In the example above, if ASBH is chosen for symbolization, the color coding will indicate the rank for ASBH, despite how the crossing ranks for alewife.) The user can choose the symbolization species with the dropdown at the top of the panel, or by hitting the symbolization icon next to the individual species. At any point, the user can also use the dropdown menu to select 'none' for symbolization, and the points selected will be gray. To remove or reset tier filters, the user can press each tier button individually or use the 'Clear All Tiers' button.

To visualize relevant habitat and other datasets concurrently with prioritization results, use 'Layers' tab.

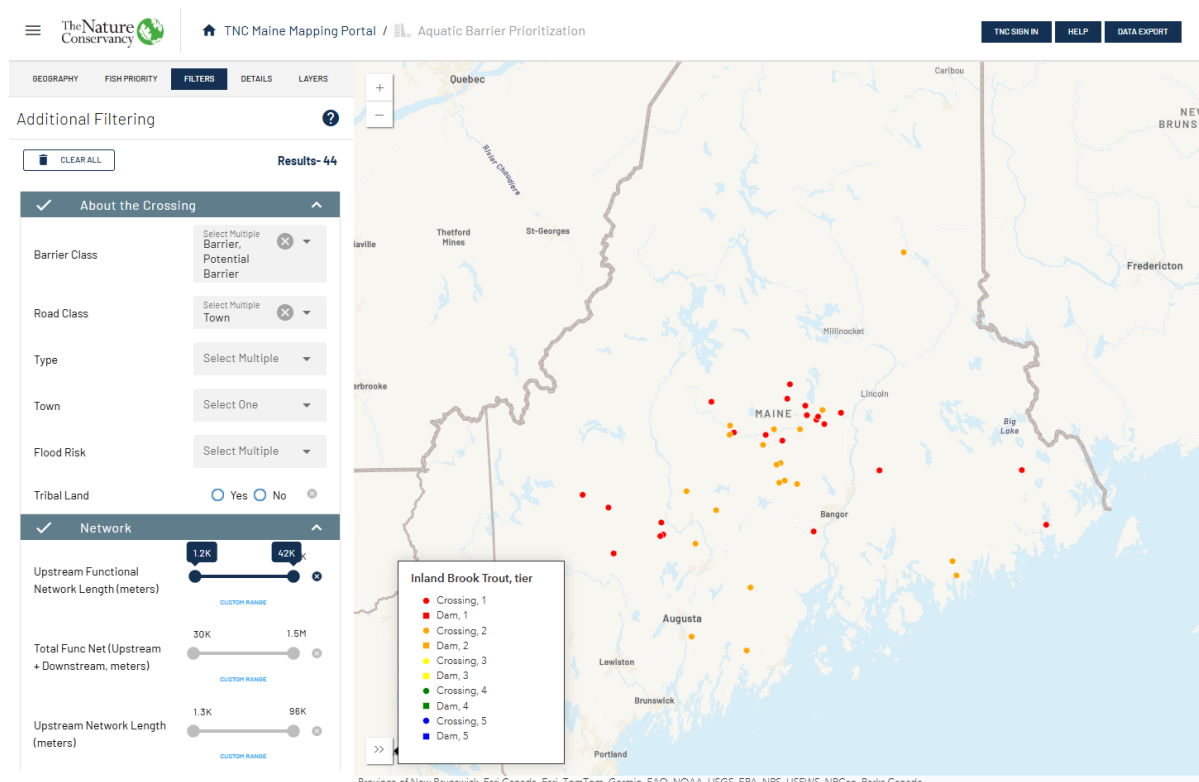


5.1.3 Filters Tab

The user can further filter within the entire dataset or their prioritized set using metrics in the eight categories provided. Numeric ranges will adjust based on the currently filtered and displayed set of crossings.

Caution! It is important to acknowledge that filtering does not change the result of prioritization, it refines the set of crossings that the user sees. Some of these metrics available for filtering are also used in the different fish prioritization scenarios (view which ones are used by clicking in the question mark by each scenario in the 'Fish Priority' tab or see Section 4.2 in this report). After selecting the tiers of prioritized results a user wants to view, additional filtering by the same metrics used in the prioritization for that species will add weight to that metric that was not anticipated when the working group (see Appendix II) determined the relative importance of each metric to the others in the prioritization.

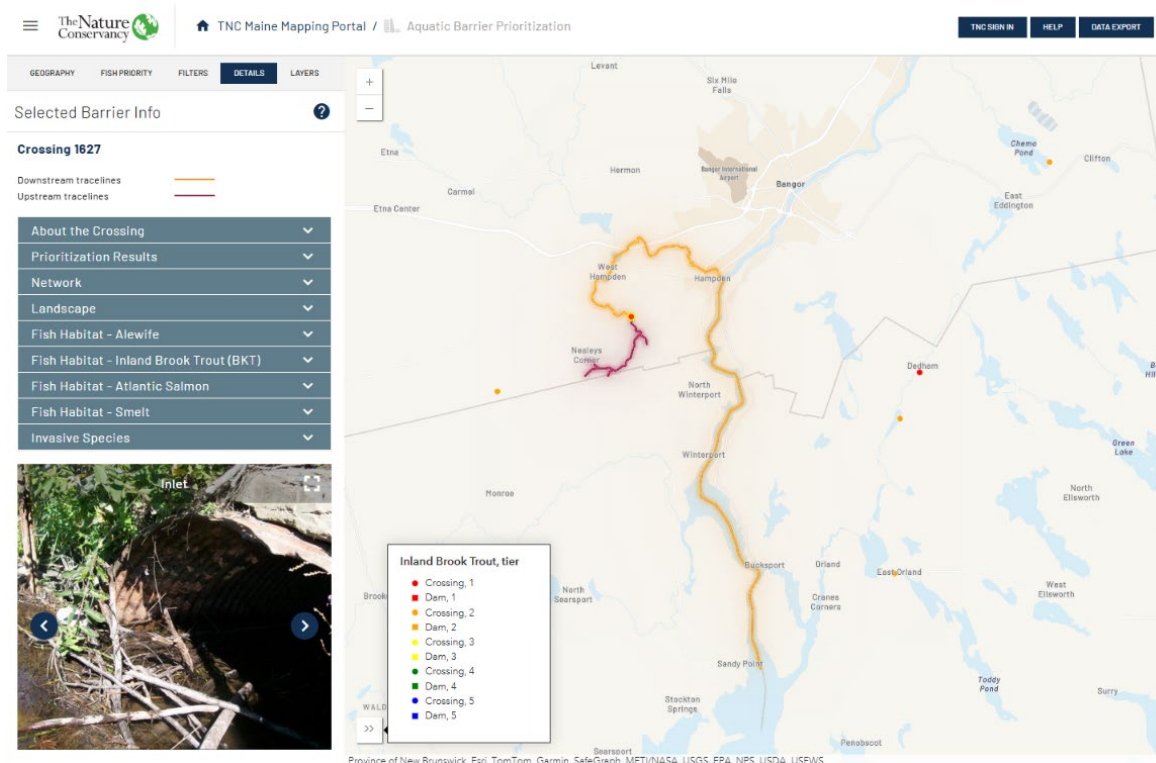
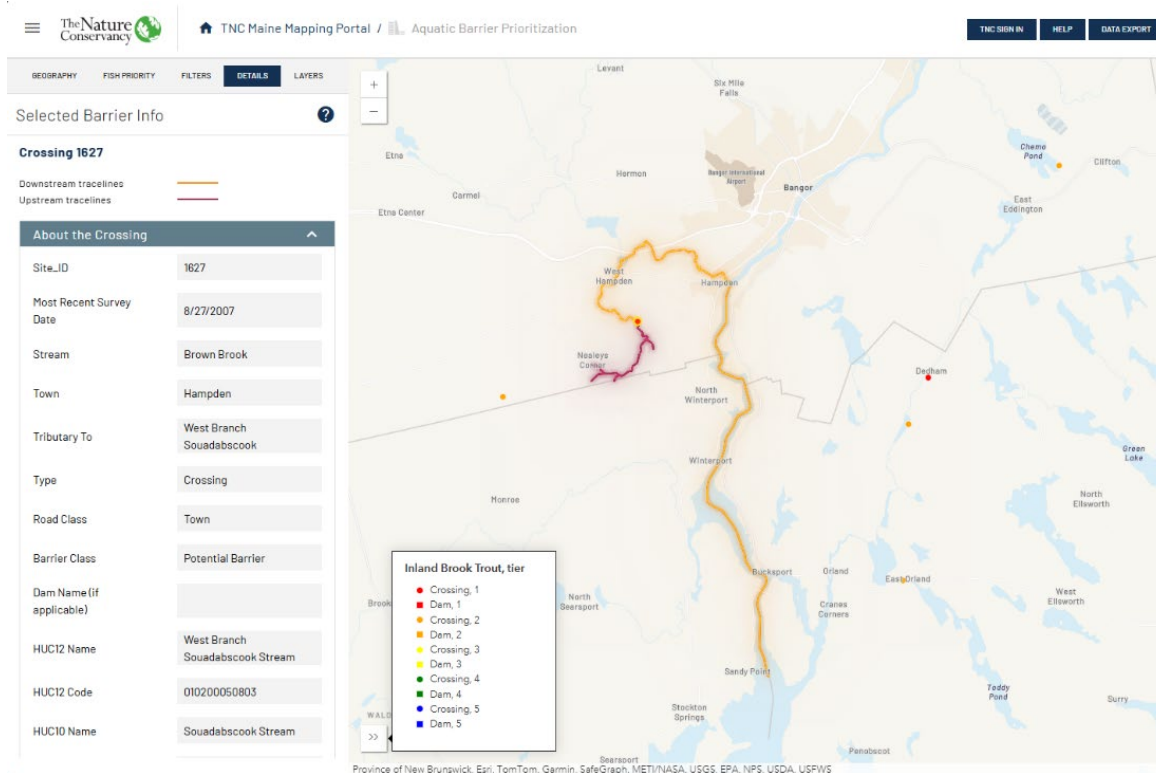
To understand each of the metrics and their contributing data sources, view the Data Sources Appendix (Appendix III) and the Glossary of Metric Definitions (Appendix IV).



5.1.4 Details Tab

When a crossing point is selected on the screen, or by searching by Site ID, the crossing will be highlighted, and the upstream and downstream network will be traced and viewable on screen. Additionally, the crossing's detailed information will be available in the eight data categories, and the photos of that crossing will be displayed.

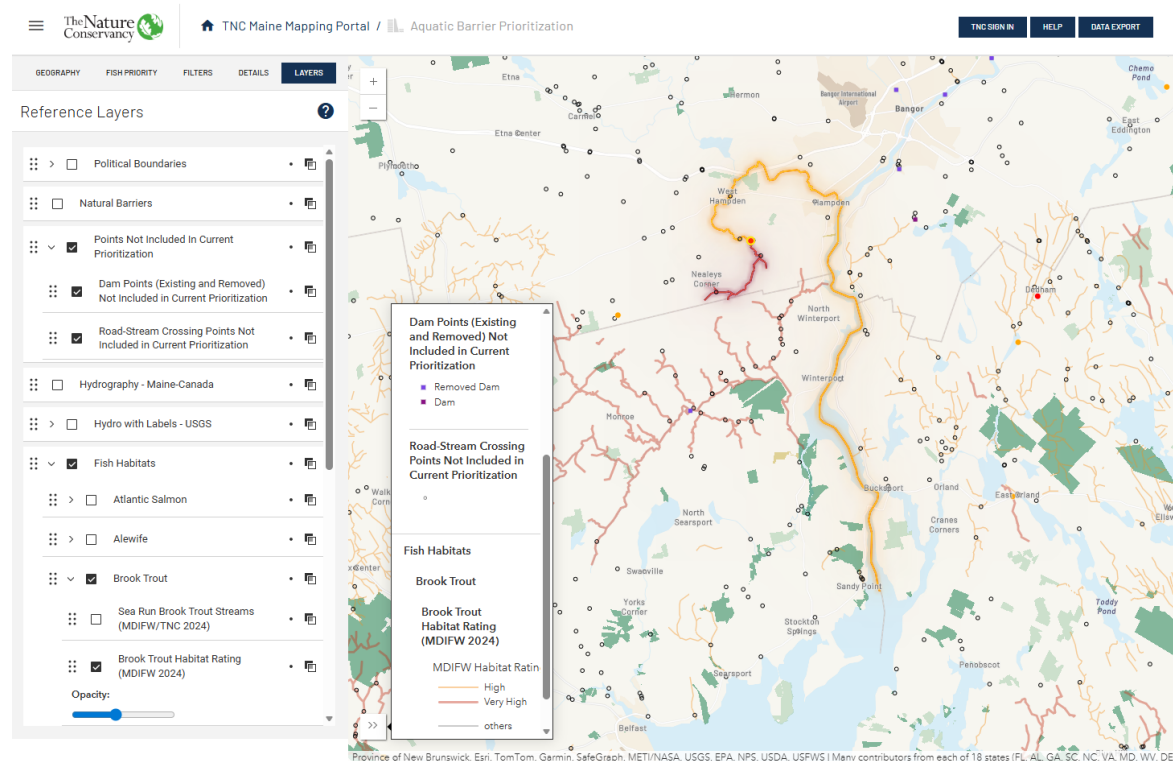
To understand each of the metrics and their contributing data sources, view the view the Data Sources Appendix (Appendix III) and the Glossary of Metric Definitions (Appendix IV).



5.1.5 Layers Tab

Turn on additional reference layers by clicking on the box next to the layer name. These layers are for display purposes only and will **not** filter or in any way impact the crossings on the screen. The user can click layers on and off by checking the boxes near the layer title; can change the order of the layers on the map by clicking on the six dots to the left of the layer and dragging into the desired position; can change the opacity of each layer by clicking on the intersecting squares to the right of the layer name.

To understand the contributing data sources, view the Data Sources Appendix (Appendix III).



6 References

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Appendix I: Organizations who hosted road-stream crossing survey teams

Acadia National Park
 Androscoggin River Watershed Council
 Atlantic Salmon Federation
 Androscoggin Valley Soil and Water Conservation District
 Belfast Bay Watershed Coalition
 Casco Bay Estuary Partnership
 Downeast Salmon Federation
 East Machias Aquatic Research Center
 Elliottsville Plantation, Inc., (now Elliottsville Foundation, Inc.)
 Houlton Band of Maliseet Indians
 Kennebec Estuary Land Trust
 Lower Penobscot Watershed Coalition
 Maine Audubon
 Maine Department of Conservation, Divisions of Parks and of Public Lands
 Maine Department of Inland Fisheries and Wildlife
 Maine Department of Marine Resources, Bureau of Sea-run Fisheries
 Maine Department of Transportation
 Maine Forest Service
 Maine Lakes Environmental Association
 NextEra Energy
 NOAA National Marine Fisheries (Maine Field Office)
 Oxford County Soil and Water Conservation District
 Passamaquoddy Tribe
 Piscataquis County Soil and Water Conservation District
 Penobscot Indian Nation
 Plum Creek (now Weyerhaeuser Company)
 Portland Water District
 Presumpscot River Watch
 Seven Islands Land Company
 Sewall Company
 Sheepscot Valley Conservation Alliance (now MidCoast Conservancy)
 Saco River Salmon Club
 Sheepscot River Watershed Council
 Stantec
 The Nature Conservancy in Maine
 Trout Unlimited

University of Maine

University of New England

USDA Natural Resource Conservation Service

US Fish and Wildlife Service (Gulf of Maine Program, Maine Refuges, *and* Maine Field Office)

US National Park Service

Wagner Forest Management

Western Foothills Land Trust

Wells National Estuarine Research Reserve

Appendix II: Original 2018 Maine Aquatic Barrier Prioritization Tool Workgroup

Name	Affiliation
Alex Abbott	Contractor to USFWS
Andy Goode	Atlantic Salmon Federation
Ben Naumann	NRCS
Ben Matthews	The Nature Conservancy in Maine
Casey Clark	Maine Department of Marine resources
Dan Kircheis	NOAA
Dan McCaw	Penobscot Indian Nation
Dan Coker	The Nature Conservancy in Maine
David Courtemanch	The Nature Conservancy in Maine
Ed Basset	Passamaquoddy at Sipayik
Eileen Bader Hall	The Nature Conservancy in Maine
Eric Ham	Maine Department of Transportation
Erik Martin	The Nature Conservancy in Maine
Gail Whipplehauser	Maine Department of Marine resources
Hadley Couraud	The Nature Conservancy in Maine
Jacob Amen	Wells National Estuarine Reserve
Jeremy Bell	The Nature Conservancy in Maine
Joshua Royte	The Nature Conservancy in Maine
Jeff Reardon	Trout Unlimited
Landis Hudson	Maine Rivers
Miranda Nemeth	Atlantic Salmon Federation
Matt Craig	Casco Bay Estuary Program
Merry Gallagher	Maine Dept. of Inland Fish & Wildlife
Molly Payne Wynne	The Nature Conservancy in Maine
Nicholas Popoff	Maine Department of Marine resources
Pat Sirois	Maine Forest Landowners
Sam Roy	University of Maine
Sean Ledwin	Maine Department of Marine resources
Sara Haggerty	Maine Audubon
Sharri Venno	Houlton Band of Maliseet Indians
Shri Verrill	Mid Coast Conservancy
Slade Moore	Maine Coastal Program

Glossary of Metrics for the Maine Aquatic Barrier Prioritization Tool

Published in April 2026

List of fish species in metrics

Common Name	Scientific Name
Alewife	<i>Alosa pseudoharengus</i>
American shad	<i>Alosa sapidissima</i>
Atlantic salmon	<i>Salmo salar</i>
Blueback herring	<i>Alosa aestivalis</i>
Eastern Brook trout	<i>Salvelinus fontinalis</i>
Rainbow smelt	<i>Osmerus mordax</i>

List of entities cited in metrics

Entity Name	Entity Acronym
Maine Department of Inland Fisheries and Wildlife	IFW
Maine Department of Marine Resources	DMR
Maine Geological Survey	MGS
Multi-Resolution Land Characteristics Consortium	MRLC Consortium
National Land Cover Database	NLCD
The Nature Conservancy, Maine	TNC Maine
National Oceanic and Atmospheric Administration	NOAA
U.S. Department of Agriculture – Natural Resources Conservation Service	USDA – NRCS
U.S. Fish and Wildlife Service	USFWS
United State Geologic Survey	USGS

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[Metrics displayed in prioritization tool]

GIS Name	Full Name	Category in Tool	Prioritization Scenarios	Slide Num.
Passability	Passability	About the Crossing	N/A	13
BatFuncUS	Upstream Functional Network Length	Network	Coastal Anadromous, Inland Brook Trout, Atlantic Salmon, American shad/Blueback herring	14
batTotUSDS	Total Functional Network Length	Network	Inland Brook Trout	15
batLenUS	Total Upstream Network Length	Network	N/A	16
batCountDS	Count of Downstream Barriers	Network	Atlantic Salmon, Alewife, American shad/Blueback herring	17
DSHydro	Count of downstream hydropower facilities	Network	American shad/Blueback herring	18
DSFishways	Count of downstream fish passage facilities	Network	Atlantic salmon, Alewife, American shad/Blueback herring	19
batAbs	Absolute Gain	Network	Inland Brook Trout	20
dsPassabiliy Product	Downstream Passability Product	Network	Coastal Anadromous, Atlantic Salmon, Alewife, American shad/Blueback herring	21
Tidal	Barrier is located on a tidal stream (current HAT +1m)	Landscape	NA	22
CoastalStreams	Barrier is located on a Coastal Stream	Landscape	NA	23
onConsLand	Barrier is on conservation land	Landscape	NA	24

Table of Contents [Metrics displayed in prioritization tool]

GIS Name	Full Name	Category in Tool	Prioritization Scenarios	Slide Num.
AquiferOr CoarseSed	Aquifer OR coarse sediments in upstream or downstream functional networks or both	Landscape	Coastal Anadromous, Inland brook trout	25
CalcModCalc	Calcareous/moderately calcareous geology in upstream or downstream networks or both	Landscape	Coastal Anadromous, Inland brook trout	26
DA_PercFor	Percent forested landcover in upstream drainage area	Landscape	Coastal Anadromous, Inland Brook Trout, Atlantic salmon, Alewife, American shad/Blueback herring	27
buff100_Tot_Perc_Statewide_forestMerged	Percent forest cover in 100m buffer of total functional network	Landscape	Coastal Anadromous, Inland Brook Trout	28
buff100_US_Perc_Statewide_forestMerged	Percent forest cover in 100m buffer of upstream network	Landscape	NA	29
usAlewifeAcres	Upstream acres of alewife ponds	Fish Habitat - Alewife	Alewife	30
occ_temp7p20	Probability of brook trout occupancy (+2degC)	Fish Habitat - Inland Brook Trout (BKT)	Inland Brook Trout	31
HeritagePondBarrier	Heritage Fish Pond Barrier	Fish Habitat - Inland Brook Trout (BKT)	Inland Brook Trout	32
MilesBT_HVH_Usonly	Number of miles of (high, very high) brook trout habitat in US network	Fish Habitat - Inland Brook Trout (BKT)	Coastal Anadromous, Inland Brook Trout	33

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[Metrics displayed in prioritization tool]

GIS Name	Full Name	Category in Tool	Prioritization Scenarios	Slide Num.
MilesBT_HVH	Number of miles of (high, very high) brook trout habitat in US + DS network	Fish Habitat - Inland Brook Trout (BKT)	Coastal Anadromous, Inland Brook Trout	34
EBTJV_Patch	EBTJV Wild Brook Trout Patch	Fish Habitat - Inland Brook Trout (BKT)	NA	35
bkt_p20_occ_gt50	Probability of brook trout occupancy >=50% (+2degC)	Fish Habitat - Inland Brook Trout (BKT)	Inland Brook Trout	36
SeaRunBT	Barrier to Sea-Run Brook Trout	Fish Habitat - Inland Brook Trout (BKT)	Coastal Anadromous	37
HUC10_CritSalmHab	Barrier is in a Salmon Critical Habitat HUC10	Fish Habitat - Atlantic Salmon	Atlantic Salmon	38
SalmonHabQual	Atlantic Salmon Habitat Quality	Fish Habitat - Atlantic Salmon	Atlantic Salmon	39
US_SalmHabUnits	Summed Salmon Habitat Units in US Network	Fish Habitat - Atlantic Salmon	Atlantic Salmon	40
HUC10_Habit_Units	Salmon Habitat Units in HUC10	Fish Habitat - Atlantic Salmon	NA	41
cumulative SalmHabUnits	Cumulative Upstream Salmon Habitat Units	Fish Habitat - Atlantic Salmon	Atlantic Salmon	42
usSmeltSite	Smelt spawning sites upstream of barrier	Fish Habitat - Smelt	Coastal Anadromous	43
InvasiveBarrier	Species of Management Concern on one side of a barrier and not the other	Invasive Species	NA	44

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[Metrics calculated, but not displayed in tool]

GIS Name	Full Name	Category in Tool	Prioritization Scenarios	Slide Num.
batFuncDS	Downstream Functional Network Length	Network	NA	46
cumulativeUSNet	Cumulative Upstream Functional Network	Network	NA	47
batCountUS	Upstream Barrier Count	Network	NA	48
batUSDnsty	Density of barriers on upstream network	Network	NA	49
DSFalls	Number of natural barriers on downstream flowpath	Network	NA	50
usNatBar	One or more of the next upstream barriers is a natural barrier	Network	NA	51
usRestPot	Upstream Restoration Potential	Network	NA	52
Aquifer	Aquifers in upstream or downstream functional networks or both	Landscape	NA	53
usCons100	Percent conserved land in 100m buffer of upstream functional network	Landscape	NA	54
dsCons100	Percent conserved land in 100m buffer of downstream functional network	Landscape	NA	55
buff100_DS_Perc_Statewide_forest Merged	Percent forest cover in 100m buffer of downstream network	Landscape	NA	56
buff100_US_Perc_nlcd_2016_developedMedHigh	Percent medium/high intensity developed landcover in 100m buffer of upstream network	Landscape	NA	57

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[Metrics calculated, but not displayed in tool]

GIS Name	Full Name	Category in Tool	Prioritization Scenarios	Slide Num.
buff100_DS_Perc_nlcd_2016_developedMedHigh	Percent medium/high intensity developed landcover in 100m buffer of downstream network	Landscape	NA	58
buff100_Tot_Perc_nlcd_2016_developedMedHigh	Percent medium/high intensity developed landcover in 100m buffer of total functional network	Landscape	NA	59
buff100_US_Perc_nlcd_2016_cultivatedCrops	Percent cultivated crops in 100m buffer of upstream network	Landscape	NA	60
buff100_DS_Perc_nlcd_2016_cultivatedCrops	Percent cultivated crops in 100m buffer of downstream network	Landscape	NA	61
buff100_Tot_Perc_nlcd_2016_cultivatedCrops	Percent cultivated crops in 100m buffer of total functional network	Landscape	NA	62
buff100_US_Perc_nlcd_2016_pastureHay	Percent pasture/hay in 100m buffer of upstream network	Landscape	NA	63
buff100_DS_Perc_nlcd_2016_pastureHay	Percent pasture/hay in 100m buffer of downstream network	Landscape	NA	64
buff100_Tot_Perc_nlcd_2016_pastureHay	Percent pasture/hay in 100m buffer of total functional network	Landscape	NA	65

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[Metrics calculated, but not displayed in tool]

GIS Name	Full Name	Category in Tool	Prioritization Scenarios	Slide Num.
floodplain_US_Perc_Statewide_forestMerged	Percent forest cover in the floodplain of upstream network	Landscape	NA	66
floodplain_DS_Perc_Statewide_forestMerged	Percent forest cover in the floodplain of downstream network	Landscape	NA	67
floodplain_Tot_Perc_Statewide_forestMerged	Percent forest cover in the floodplain of total functional network	Landscape	NA	68
floodplain_US_Perc_nlcd_2016_developedMedHigh	Percent medium/high intensity developed landcover in the floodplain of the upstream network	Landscape	NA	69
floodplain_DS_Perc_nlcd_2016_developedMedHigh	Percent medium/high intensity developed landcover in the floodplain of the downstream network	Landscape	NA	70
floodplain_Tot_Perc_nlcd_2016_developedMedHigh	Percent medium/high intensity developed landcover in the floodplain of the total functional network	Landscape	NA	71
floodplain_US_Perc_nlcd_2016_cultivatedCrops	Percent cultivated crops in the floodplain of upstream network	Landscape	NA	72
floodplain_DS_Perc_nlcd_2016_cultivatedCrops	Percent cultivated crops in the floodplain of downstream network	Landscape	NA	73

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[Metrics calculated, but not displayed in tool]

GIS Name	Full Name	Category in Tool	Prioritization Scenarios	Slide Num.
floodplain_Tot_Perc_nlcd_2016_cultivatedCrops	Percent cultivated crops in the floodplain of total functional network	Landscape	NA	74
floodplain_US_Perc_nlcd_2016_pastureHay	Percent pasture/hay in the floodplain of upstream network	Landscape	NA	75
floodplain_DS_Perc_nlcd_2016_pastureHay	Percent pasture/hay in the floodplain of downstream network	Landscape	NA	76
floodplain_Tot_Perc_nlcd_2016_pastureHay	Percent pasture/hay in the floodplain of total functional network	Landscape	NA	77
NESZCL_Int	River Size Class	Landscape	NA	78
MilesUS_HATcurrent	Miles of tidal habitat (current) upstream of barrier	Landscape/Fish Habitat	NA	79
MilesUS_HATPlus1m	Miles of tidal habitat (current HAT + 1m) upstream of barrier	Landscape/Fish Habitat	NA	80
MilesUS_HATPlus6ft	Miles of tidal habitat (current HAT + 6ft) upstream of barrier	Landscape/Fish Habitat	NA	81

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[Metrics calculated, but not displayed in tool]

GIS Name	Full Name	Category in Tool	Prioritization Scenarios	Slide Num.
bkt_curr_occ_gt50	Probability of brook trout occupancy >=50% (current temps)	Fish Habitat - Inland Brook Trout (BKT)	NA	82
BT_1_4_HVH	High or Very High quality brook trout habitat upstream OR downstream of barrier	Fish Habitat - Inland Brook Trout (BKT)	Inland Brook Trout	83
BT_1_4_VH	Very High quality brook trout habitat upstream OR downstream of barrier	Fish Habitat - Inland Brook Trout (BKT)	NA	84
occ_current	Probability of brook trout occupancy (current temps)	Fish Habitat - Inland Brook Trout (BKT)	NA	85
occ_temp7p40	Probability of brook trout occupancy (+4degC)	Fish Habitat - Inland Brook Trout (BKT)	NA	86
bkt_p40_occ_gt50	Probability of brook trout occupancy >=50% (+4degC)	Fish Habitat - Inland Brook Trout (BKT)	NA	87
US_SalmParrProd	Summed salmon parr productivity in US Functional Network	Fish Habitat - Atlantic Salmon	NA	88
InvasiveBarrierConfirmed	Species of Management Concern on one side of a barrier and not the other - Confirmed only	Invasive Species	NA	89
InvasiveBarrierConfNumSpp	Number of Species of Management Concern that are blocked - Confirmed only	Invasive Species	NA	90

Metrics displayed in the Aquatic Barrier Prioritization Tool

Passability

GIS Field Name: Passability

Category: About the Crossing

A numerical score from 0-1 that indicates how passable a barrier is likely to be, where 0=complete barrier and 1=fully passable. All dams are considered complete barriers, unless there is known fish passage facilities at a dam. Dams with fish passage were reviewed by J. Royte and E. Martin from The Nature Conservancy and assigned a passability score based on the efficacy of passage at the dam. Passability for road-stream crossings is first assigned based on Maine's crossing assessment protocol which splits crossings into barriers, potential barriers, and no barrier. Barriers were assigned a value of 0 and no barriers were assigned a value of 1.

Potential barriers were further refined based on Froude number which is a proxy for the relationship between velocity and depth that can be utilized as a screening metric to identify crossings that have a high probability of being a velocity barrier to fish passage. Passability score of 0.66 correlates directly to a Froude number of 1, so only potential barriers with a score of < 0.66 were prioritized in the final result.

This metric is used in the generation of other metrics.

Unit: Index

Upstream Functional Network Length

GIS Field Name: batFuncUS

Category: Network

Used in Prioritization Scenario(s): Coastal Anadromous, Inland Brook Trout, Atlantic salmon, American shad/Blueback herring

Length of the functional network upstream of a barrier.

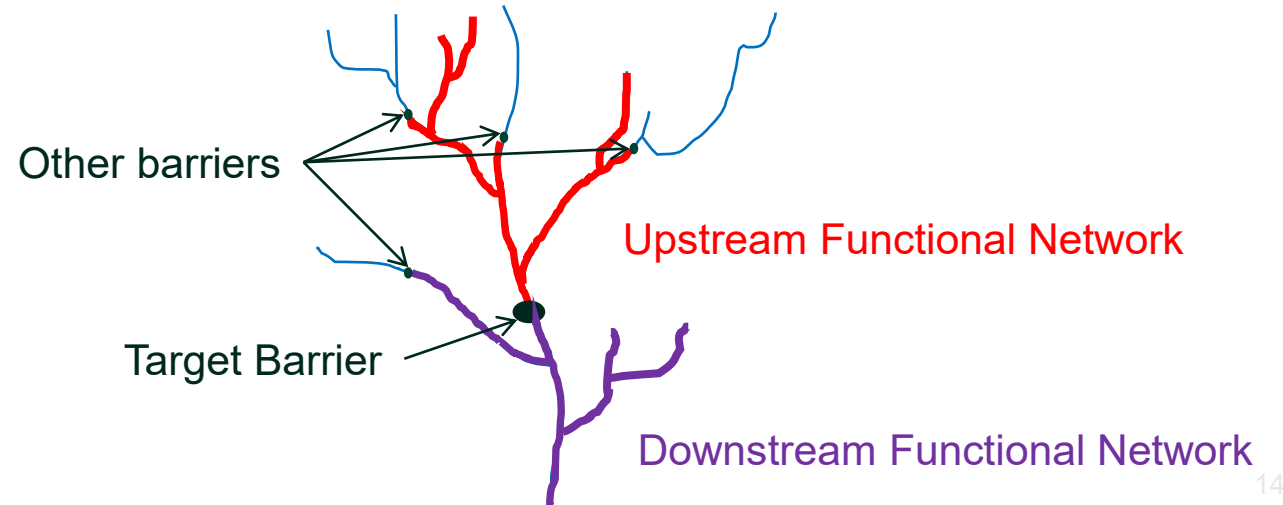
“Functional network” is defined by those sections of river that a fish could theoretically access from any other point within that functional network. Terminal ends are barriers, headwaters, and/or river mouth.

Unit: meters (converted to miles in some outputs)

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine



Total Functional Network Length

GIS Field Name: batTotUSDS

Category: Network

Used in Prioritization Scenario(s): Inland Brook Trout

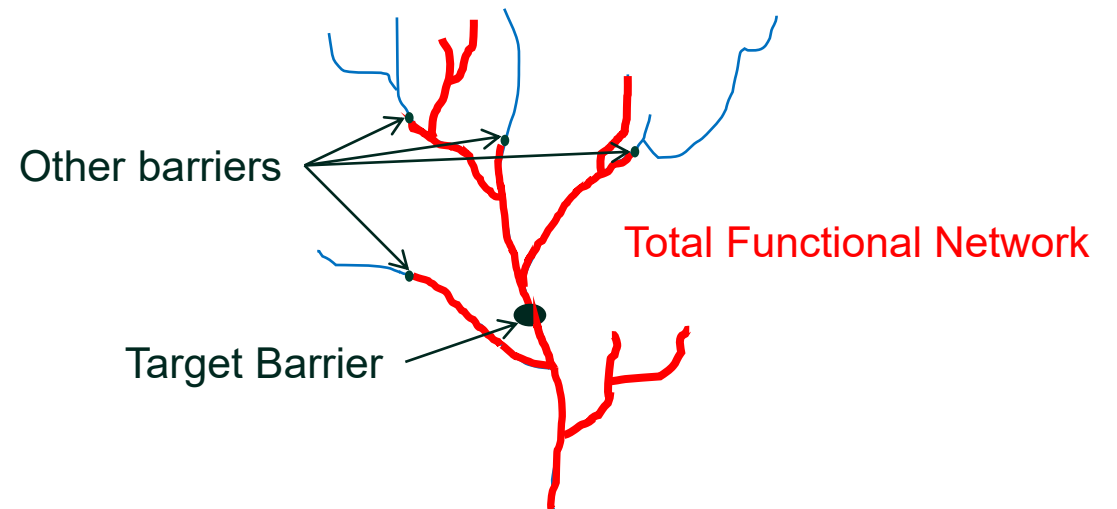
Length of the functional network downstream of a barrier plus the length of the functional network upstream of a barrier. This is the length of the recombined network if the barrier was removed.

Unit: meters (converted to miles in some outputs)

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine



Total Upstream Network Length

GIS Field Name: batLenUS

Category: Network

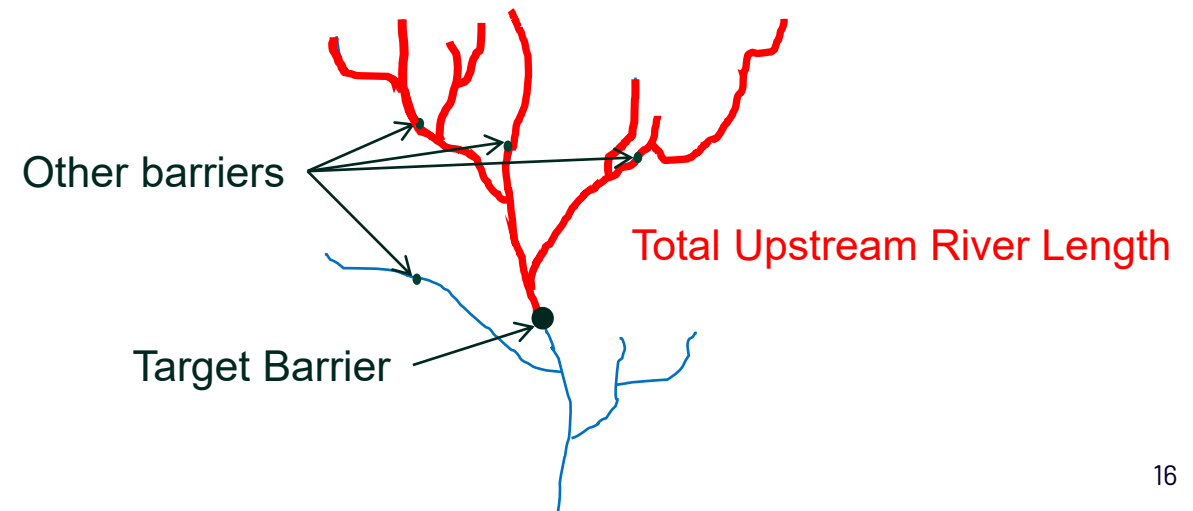
Total length of river network upstream of a given barrier, regardless of any upstream barriers.

Unit: meters

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine



Count of Downstream Barriers

GIS Field Name: batCountDS

Category: Network

Used in Prioritization Scenario(s): Atlantic salmon, Alewife, American shad/Blueback herring

The number of barriers downstream of a given barrier. Includes natural barriers (waterfalls), which are included in network dataset creation

Does not include barriers excluded from network dataset creation

Unit: #

Sort Order: Ascending

Data Sources:

- Hydrography (2023): USGS, TNC Maine

Count of downstream hydropower facilities

GIS Field Name: DSHydro

Category: Network

Used in Prioritization Scenario(s): American shad/Blueback herring

The number of hydropower facilities downstream of a barrier

Unit: #

Sort Order: Ascending

Data Sources:

- Hydrography (2023): USGS, TNC Maine
- Dam information (2025): TNC Maine compiled

Count of downstream fish passage facilities

GIS Field Name: DSFishways

Category: Network

Used in Prioritization Scenario(s): Atlantic salmon, Alewife, American shad/Blueback herring

The number of fish passage facilities downstream of a barrier

Unit: #

Sort Order: Descending

Data Sources:

- Hydrography (2023), USGS, TNC Maine
- Fishway info: DMR (2024)

Absolute Gain

GIS Field Name: batAbs

Category: Network

Used in Prioritization Scenario(s): Inland Brook Trout

This metric is the minimum of the two functional networks of a barrier. For example, if the upstream functional network is 10 kilometers and downstream functional network is 5 kilometers, then the Absolute Gain is 5 kilometers.

Unit: meters

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine

Downstream Passability Product

GIS Field Name: dsPassabilityProduct

Category: Network

Used in Prioritization Scenario(s): Coastal Anadromous, Atlantic salmon, Alewife, American shad/Blueback herring

The product (multiplication) of all passability scores on the downstream flow path of a barrier.

Unit: Index

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine

Barrier is located on a tidal stream (current HAT +1m)

GIS Field Name: Tidal

Category: Landscape

Barrier is located on a tidal stream, based on the spatial intersection of the barrier and the Highest Annual Tide (HAT2018) plus 1m dataset:

Unit: Boolean

Sort Order: Descending

Data Sources:

- Tidal Waters / Sea-level Rise Scenarios (2018): MGS

Barrier is Located on a Coastal Stream

GIS Field Name: CoastalStreams

Category: Landscape

Barrier is located on a small coastal stream – Small Coastal streams were defined as streams with an upstream drainage area less than 100 km² that drain into a tidal area (directly into the ocean or the tidal portion of a larger river).

Unit: Boolean

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine
- Tidal Waters / Sea-level Rise Scenarios (2018): MGS

Barrier is on conservation land

GIS Name: onConsLand

Category: Landscape

Conservation Lands included in the current analysis included those with gap status 1,2,3, and 39 – based on TNC's modified version of USGS's [Gap Status Codes](#).

Unit: Boolean

Data Sources:

- Conservation Lands / Secured Lands (2025): TNC Maine

Aquifer OR coarse sediments in upstream or downstream functional networks or both

GIS Field Name: AquiferOrCoarseSed

Category: Landscape

Used in Prioritization Scenario(s): Coastal Anadromous, Inland Brook Trout

[Aquifers](#) OR coarse sediments are found in [upstream functional network](#) OR [downstream functional network](#) OR both. This metric evaluates the potential for cold water habitats within a barrier's respective networks.

Unit: Boolean

Sort Order: Descending

Data Sources:

- Aquifers (2020): MGS
- Coarse Sediments (2006): The Nature Conservancy Ecological Land Units

Calcareous / moderately calcareous geology in upstream or downstream networks or both

GIS Field Name: CalcModCalc

Category: Landscape

Used in Prioritization Scenario(s): Coastal Anadromous, Inland Brook Trout

Calcareous or moderately calcareous geology is found in [upstream functional network](#) OR [downstream functional network](#) OR both. This metric evaluates the acid buffering capacity within a barrier's respective networks.

Unit: Boolean

Sort Order: Descending

Data Sources:

- Calcareous/Moderately Calcareous Bedrock (2006): The Nature Conservancy Ecological Land Units

Percent forested landcover in upstream drainage area

GIS Name: DA_PercFor

Category: Landscape

Used in Prioritization Scenario(s): Coastal Anadromous, Inland Brook Trout, Atlantic salmon, Alewife, American shad/Blueback herring

Percent forested landcover in entire upstream watershed.

Forested landcover classes were aggregated separately for northern Maine and southern Maine. In northern Maine, forest cover was defined as deciduous forest, evergreen forest, mixed forest, and woody wetlands. In southern Maine, scrub/shrub was also included as a forest class based on input from Merry Gallagher, Maine DIFW.

Unit: %

Sort Order: Descending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent forest cover in 100m buffer of total functional network

GIS Name: buff100_Tot_Perc_Statewide_forestMerged

Category: Landscape

Used in Prioritization Scenario(s): Coastal Anadromous, Inland Brook Trout

Percent forested landcover in a 100m buffer of a barrier's total functional network (upstream + downstream).

Forested landcover classes were aggregated separately for northern Maine and southern Maine. In northern Maine, forest cover was defined as deciduous forest, evergreen forest, mixed forest, and woody wetlands. In southern Maine, scrub/shrub was also included as a forest class based on input from Merry Gallagher, Maine IFW.

Unit: %

Sort Order: Descending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent forest cover in 100m buffer of upstream network

GIS Name: buff100_US_Perc_Statewide_forestMerged

Category: Landscape

Percent forested landcover in a 100m buffer of a barrier's upstream functional network.

Forested landcover classes were aggregated separately for northern Maine and southern Maine. In northern Maine, forest cover was defined as deciduous forest, evergreen forest, mixed forest, and woody wetlands. In southern Maine, scrub/shrub was also included as a forest class based on input from Merry Gallagher, Maine DIFW.

Unit: %

Sort Order: Descending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Upstream acres of alewife ponds

GIS Field Name: usAlewifeAcres

Category: Fish Habitat – Alewife

Used in Prioritization Scenario(s): Alewife

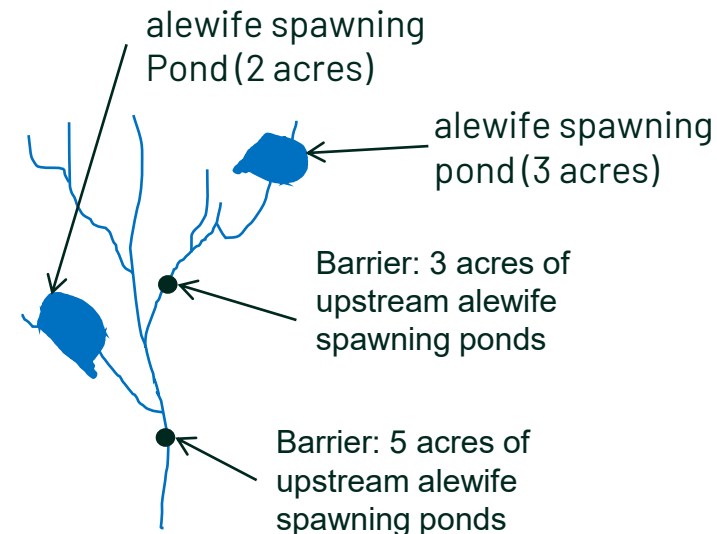
The summed acreage of documented alewife spawning ponds upstream of the barrier. This metric evaluates potential upstream alewife spawning habitat, thus ignoring any upstream barriers.

Unit: Acres

Sort Order: Descending

Data Sources:

- Alewife Ponds (2024): DMR, TNC Maine



Probability of brook trout occupancy (+2°C)

GIS Field Name: occ_temp7p20

Category: Fish Habitat – Inland Brook Trout (BKT)

Used in Prioritization Scenario(s): Inland Brook Trout

Occupancy probability of Eastern Brook trout for the catchment (NHDHRDV2) that the barrier is located within, given a 2°C increase in July water temperature.

Unit: %

Sort Order: Descending

Data Sources:

- EcoSheds Catchments (2024): USGS/EcoSheds program

Heritage Fish Pond Barrier

GIS Field Name: HeritagePondBarrier

Category: Fish Habitat – Inland Brook Trout (BKT)

Used in Prioritization Scenario(s): Inland Brook Trout

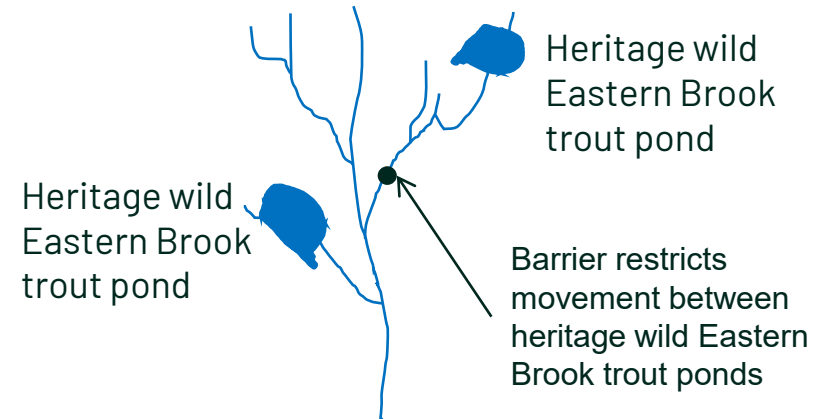
Barrier separates two functional network that both include heritage wild Eastern Brook trout ponds.

Unit: Boolean

Sort Order: Descending

Data Sources:

- Maine Heritage Fish Waters (2025): IFW



Number of miles of (high, very high) brook trout habitat in US network

GIS Field Name: MilesBT_HVH_USonly

Category: Fish Habitat – Inland Brook Trout (BKT)

Used in Prioritization Scenario(s): Coastal Anadromous, Inland Brook Trout

Length of Eastern Brook trout habitat in the Upstream Functional Network that is of high or very high priority for Eastern Brook trout

Unit: Miles

Sort Order: Descending

Data Sources:

- Brook Trout Habitat Rating (2024): IFW

Number of miles of (high, very high) brook trout habitat in US + DS networks

GIS Field Name: MilesBT_HVH

Category: Fish Habitat – Inland Brook Trout (BKT)

Used in Prioritization Scenario(s): Coastal Anadromous, Inland Brook Trout

Length of Eastern Brook trout habitat in the combined [Upstream Functional Network](#) + [Downstream Functional Network](#) that is of high, or very high priority for Eastern Brook trout

Unit: Miles

Sort Order: Descending

Data Sources:

- Brook Trout Habitat Rating (2024): IFW

EBTJV Wild Brook Trout Patch

GIS Field Name: EBTJV_Patch

Category in Tool: Fish Habitat – Inland Brook Trout (BKT)

Unit: Boolean

Sort Order: Descending

Data Sources:

- Eastern Brook Trout Joint Venture Wild Trout Patches (2025): Eastern Brook Trout Joint Venture

Probability of brook trout occupancy $\geq 50\%$ ($+2^{\circ}\text{C}$)

GIS Field Name: bkt_p20_occ_gt50

Category: Fish Habitat - Inland Brook Trout (BKT)

Used in Prioritization Scenario(s): Inland Brook Trout

Occupancy probability of Eastern Brook trout for the catchment (NHDHRDV2) that the barrier is located within is $\geq 50\%$, given a 2°C increase in July water temperature.

Unit: Boolean

Sort Order: Descending

Data Sources:

- EcoSheds Catchments (2024): USGS/EcoSheds program

Barrier to Sea-Run Brook Trout

GIS Field Name: SeaRunBT

Category: Fish Habitat – Inland Brook Trout (BKT)

Used in Prioritization Scenario(s): Coastal Anadromous

Barrier potentially blocks sea-run Eastern Brook trout migrations because it falls on reaches mapped for sea-run Eastern Brook trout use.

Unit: Boolean

Sort Order: Descending

Data Sources:

- Sea Run Brook Trout Streams (2024): TNC Maine, IFW

Barrier is in a Salmon Critical Habitat HUC10

GIS Field Name: HUC10_CritSalmHab

Category: Fish Habitat – Atlantic Salmon

Used in Prioritization Scenario(s): Atlantic salmon

Barrier is located within a designated Atlantic salmon Critical Habitat [10-Digit Hydrologic Unit](#).

Unit: Boolean

Sort Order: Descending

Data Sources:

- Atlantic Salmon Critical Habitat (2023): USDA-NRCS, NOAA

Atlantic Salmon Habitat Quality

GIS Name: SalmonHabQual

Category: Fish Habitat – Atlantic Salmon

Used in Prioritization Scenario(s): Atlantic salmon

The Atlantic salmon habitat quality ranking of the flowline on which the barrier falls.

Unit: Categorical (Very High (4), High (3), Medium(2), Low(1), No (0))

Sort Order: Descending

Data Sources:

- Atlantic Salmon Habitat Quality (2024), DMR, TNC Maine

Summed Salmon Habitat Units in US Network

GIS Field Name: US_SalmHabUnits

Category: Fish Habitat – Atlantic Salmon

The summed value of Atlantic salmon habitat units obtained from a GIS-based habitat model that predicts the amount of accessible Atlantic salmon rearing habitat throughout the Gulf of Maine Atlantic salmon DPS

Units: Habitat units (sq. meters/100)

Sort Order: Descending

Data Sources:

- Atlantic Salmon Modeled Habitat Units (2012): USFWS Gulf of Maine Coastal Program, NOAA

Salmon Habitat Units in HUC10

GIS Field Name: HUC10_Habitat_Units

Category: Fish Habitat – Atlantic Salmon

The sum of Atlantic salmon habitat units within the 10-Digit Hydrologic Unit (HUC10) that the barrier is within

Unit: Index

Sort Order: Descending

Data Sources:

- Atlantic Salmon Modeled Habitat Units (2012): USFWS Gulf of ME Coastal Program, NOAA

Cumulative Upstream Salmon Habitat Units

GIS Field Name: cumulativeSalmHabUnits

Category: Fish Habitat – Atlantic Salmon

Used in Prioritization Scenario(s): Atlantic salmon

The cumulative upstream Atlantic salmon habitat units metric assesses the amount of upstream Atlantic salmon habitat units that would be made available upstream of a barrier, accounting accounts for the cumulative impacts of partial passability at upstream barriers. For a given target barrier, it is calculated as the sum of [the probability weighted salmon habitat units](#) in the barrier’s upstream functional network, plus the weighted Atlantic salmon habitat units in the upstream functional networks of all of its upstream barriers multiplied by the cumulative [passability score](#) between the target barrier and each upstream barrier. Given the sketch below, the cumulative upstream Atlantic salmon habitat units for the target barrier would be:

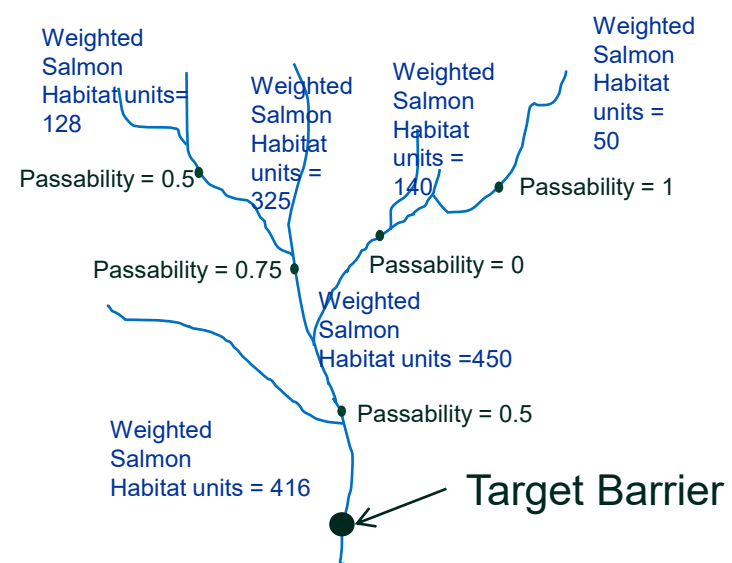
$$\begin{aligned}
 & (416)+(450*0.5)+(325*0.5*0.75)+(128*0.5*0.75*0.5)+(140*0.5*0)+(50*0.5*0*1) \\
 & = \\
 & 416 + 225 + 121.875 + 24 + 0 + 0 \\
 & = \\
 & \text{786.875}
 \end{aligned}$$

Unit: Index (cumulative salmon habitat * cumulative passability)

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine
- Atlantic Salmon Modeled Habitat Units (2012): USFWS Gulf of ME Coastal Program, NOAA



Smelt spawning sites upstream of barrier

GIS Field Name: usSmeltSite

Category: Fish Habitat – Smelt

Used in Prioritization Scenario(s): Coastal Anadromous

Value of smelt spawning sites priorities located upstream of barrier. Priority classes were assigned the numerical values in the table below. If more than one spawning site is located above the barrier, the maximum value is assigned to the barrier.

Unit: Boolean (Yes/No)

Sort Order: Descending

Data Sources:

- Smelt Reaches (2024): DMR, TNC Maine

Species of Management Concern on one side of a barrier and not the other

GIS Field Name: InvasiveBarrier

Category: Invasive Species

Species of management concern are found on one side of a barrier and not the other. Species evaluated include Black crappie, Largemouth bass, Muskellunge, Northern pike, Smallmouth bass, and Walleye.

***Note!** *This dataset was not updated for the 2025 iteration of the prioritization tool.*

Unit: Boolean

Sort Order: Ascending

Data Sources:

- Fish Species of Management Concern / Invasive Fish Species (2018 approx.): IFW

**Additional metrics
calculated, but not
displayed in the tool**

Downstream Functional Network Length

GIS Field Name: batFuncDS

Category (not displayed in tool): Network

Length of the functional network downstream of a barrier.

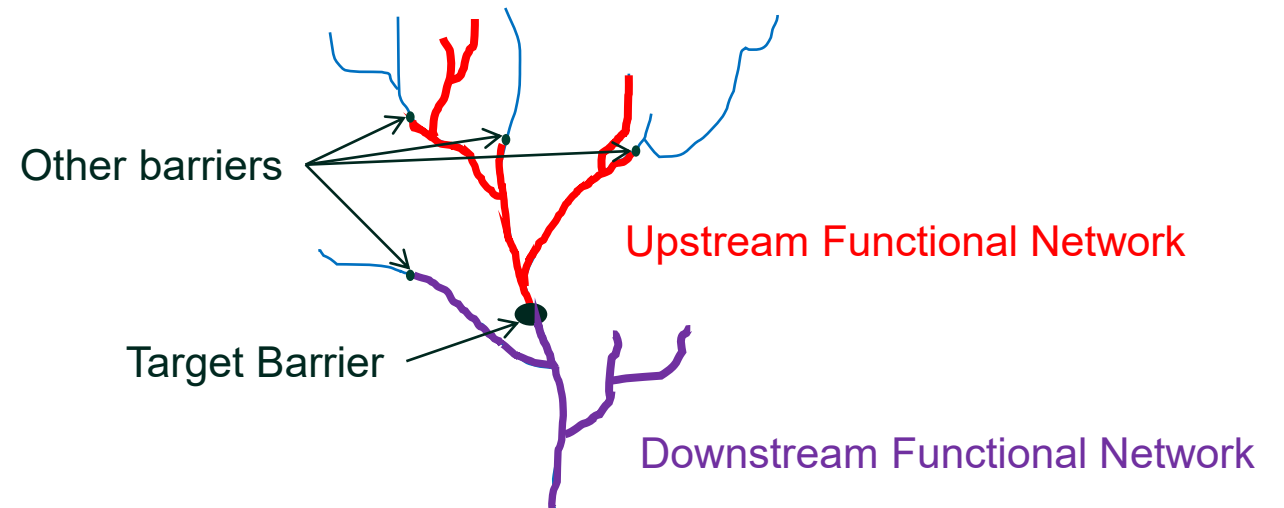
“Functional network” is defined by those sections of river that a fish could theoretically access from any other point within that functional network. Terminal ends are barriers, headwaters, and/or the river mouth.

Unit: meters (converted to miles in some outputs)

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine



Cumulative Upstream Functional Network

GIS Field Name: cumulativeUSNet

Category (not displayed in tool): Network

The cumulative upstream functional network metric assesses the amount of river habitat that would be made available upstream of a barrier, accounting accounts for the cumulative impacts of partial passability at upstream barriers. be made available upstream of a barrier, accounting accounts for the cumulative impacts of partial passability at upstream barriers. For a given target barrier, it is calculated as the sum of the barrier's upstream functional network, plus the upstream functional networks of all of its upstream barriers multiplied by the cumulative passability score between the target barrier and each upstream barrier. Given the sketch below, the cumulative upstream functional network for the target barrier would be:

$$(4,230)+(3,150*0.5)+(2,890*0.5*0.75)+(1,020*0.5*0.75*0.5)+(2,680*0.5*0)+(1,785*0.5*0*1)$$

=

$$4,230 + 1,575 + 1083.75 + 191.25 + 0 + 0$$

=

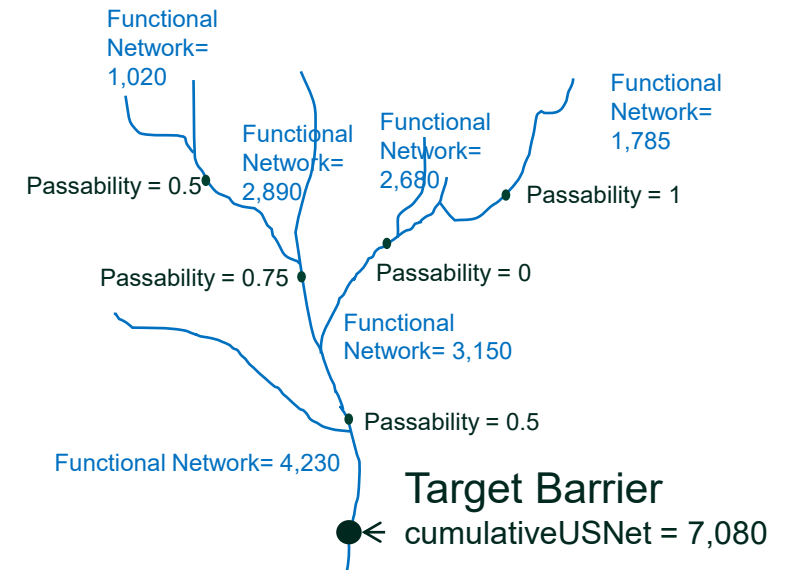
7,080

Unit: Index (cumulative meters * cumulative passability)

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine



Upstream Barrier Count

GIS Field Name: batCountUS

Category (not displayed in tool): Network

The number of barriers upstream of a given barrier.

This metric is used in the generation of other metrics. It is not available for prioritizations. Includes all barriers which are used in network generation (including natural barriers and restricted barriers).

Does not include barriers excluded from network dataset creation

Unit: #

Sort Order: Ascending

Data Sources:

- Hydrography (2023): USGS, TNC Maine

Density of barriers on upstream network

GIS Field Name: batUSDnsty

Category (not displayed in tool): Network

The number of other barriers upstream of a given barrier divided by the length of upstream river (batCountUS / batLenUS).

Includes all barriers which are used in network generation (including natural barriers and restricted barriers)

Unit: #/m

Sort Order: Ascending

Data Sources:

- Hydrography (2023): USGS, TNC Maine

Number of Natural Barriers on Downstream Flowpath

GIS Field Name: DSFalls

Category (not displayed in tool): Network

Count of natural barriers (e.g. waterfalls) on downstream flowpath of a barrier

Unit: #

Sort Order: Ascending

Data Sources:

- Hydrography (2023): USGS, TNC Maine
- Natural Barriers (2025), TNC Maine

One or More of the Next Upstream Barriers is a Natural Barrier

GIS Field Name: usNatBar

Category (not displayed in tool): Network

Value is 'Yes' if one or more of the next upstream barriers is a natural barrier (e.g. waterfall)

Unit: Boolean

Sort Order: Ascending

Data Sources:

- Hydrography (2023): USGS, TNC Maine
- Natural Barriers (2025), TNC Maine

Upstream Restoration Potential

GIS Field Name: usRestPot

Category (not displayed in tool): Network

The upstream restoration potential is an index designed to summarize the restoration potential of a barrier. It is a measure of the total upstream river habitat, divided by the number of upstream barriers, and multiplied by the passability scores of downstream barriers.

Specifically, it is calculated as

$$((((\text{batLenUS} - \text{batFuncUS}) / \text{batCountUS}) + \text{batFunUS}) * \text{dsPasabilityProduct})$$

Unit: Index

Sort Order: Descending

Data Sources:

- Hydrography (2023): USGS, TNC Maine

Aquifers in upstream or downstream functional networks or both

GIS Field Name: Aquifer

Category (not displayed in tool): Landscape

Aquifers are found in upstream functional network OR downstream functional network OR both. This metric evaluates the potential for cold water habitats within a barrier's respective networks.

Unit: Boolean

Sort Order: Descending

Data Sources:

- Aquifers (2020): MGS

Percent conserved land in 100m buffer of upstream functional network

GIS Name: usCons100

Category (not displayed in tool): Landscape

The percent of a 100m buffer of a barrier's upstream functional network that is in conservation. Conservation Lands included in the current analysis included those with gap status 1,2,3, and 39 – based on TNC's modified version of USGS's [Gap Status Codes](#).

Unit: %

Data Sources:

- Conservation Lands / Secured Lands (2025): TNC Maine

Percent conserved land in 100m buffer of downstream functional network

GIS Name: dsCons100

Category (not displayed in tool): Landscape

The percent of a 100m buffer of a barrier's downstream functional network that is in conservation. Conservation Lands included in the current analysis included those with gap status 1,2,3, and 39 – based on TNC's modified version of USGS's [Gap Status Codes](#).

Unit: %

Data Sources:

- Conservation Lands / Secured Lands (2025): TNC Maine

Percent forest cover in 100m buffer of downstream network

GIS Name: buff100_DS_Perc_Statewide_forestMerged

Category (not displayed in tool): Landscape

Percent forested landcover in a 100m buffer of a barrier's downstream functional network.

Forested landcover classes were aggregated separately for northern Maine and southern Maine. In northern Maine, forest cover was defined as deciduous forest, evergreen forest, mixed forest, and woody wetlands. In southern Maine, scrub/shrub was also included as a forest class based on input from Merry Gallagher, Maine IFW.

Unit: %

Sort Order: Descending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent medium/high intensity developed landcover in 100m buffer of upstream network

GIS Name: buff100_US_Perc_nlcd_2016_developedMedHigh

Category (not displayed in tool): Landscape

Percent of medium or high intensity developed landcover in a 100m buffer of a barrier's upstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent medium/high intensity developed landcover in 100m buffer of downstream network

GIS Name: buff100_DS_Perc_nlcd_2016_developedMedHigh

Category (not displayed in tool): Landscape

Percent of medium or high intensity developed landcover in a 100m buffer of a barrier's downstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent medium/high intensity developed landcover in 100m buffer of total functional network

GIS Name: buff100_Tot_Perc_nlcd_2016_developedMedHigh

Category (not displayed in tool): Landscape

Percent of medium or high intensity developed landcover in a 100m buffer of a barrier's total functional network (upstream + downstream).

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent cultivated crops in 100m buffer of upstream network

GIS Name: buff100_US_Perc_nlcd_2016_cultivatedCrops

Category (not displayed in tool): Landscape

Percent of cultivated crop landcover in a 100m buffer of a barrier's upstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent cultivated crops in 100m buffer of downstream network

GIS Name: buff100_DS_Perc_nlcd_2016_cultivatedCrops

Category (not displayed in tool): Landscape

Percent of cultivated crop landcover in a 100m buffer of a barrier's downstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent cultivated crops in 100m buffer of total functional network

GIS Name: buff100_Tot_Perc_nlcd_2016_cultivatedCrops

Category (not displayed in tool): Landscape

Percent of cultivated crop landcover in a 100m buffer of a barrier's total (upstream + downstream) functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent pasture/hay in 100m buffer of upstream network

GIS Name: buff100_US_Perc_nlcd_2016_pastureHay

Category (not displayed in tool): Landscape

Percent of pasture/hay landcover in a 100m buffer of a barrier's upstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent pasture/hay in 100m buffer of downstream network

GIS Name: buff100_DS_Perc_nlcd_2016_pastureHay

Category (not displayed in tool): Landscape

Percent of pasture/hay landcover in a 100m buffer of a barrier's downstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent pasture/hay in a 100m buffer of total functional network

GIS Name: buff100_Tot_Perc_nlcd_2016_pastureHay

Category (not displayed in tool): Landscape

Percent of pasture/hay landcover in a 100m buffer of a barrier's total (upstream + downstream) functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium

Percent forest cover in the floodplain of upstream network

GIS Name: floodplain_US_Perc_Statewide_forestMerged

Category (not displayed in tool): Landscape

Percent forested landcover in the floodplain of a barrier's upstream functional network.

Forested landcover classes were aggregated separately for northern Maine and southern Maine. In northern Maine, forest cover was defined as deciduous forest, evergreen forest, mixed forest, and woody wetlands. In southern Maine, scrub/shrub was also included as a forest class based on input from Merry Gallagher, Maine IF & W.

Unit: %

Sort Order: Descending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent forest cover in the floodplain of downstream network

GIS Name: floodplain_DS_Perc_Statewide_forestMerged

Category (not displayed in tool): Landscape

Percent forested landcover in the floodplain of a barrier's downstream functional network.

Forested landcover classes were aggregated separately for northern Maine and southern Maine. In northern Maine, forest cover was defined as deciduous forest, evergreen forest, mixed forest, and woody wetlands. In southern Maine, scrub/shrub was also included as a forest class based on input from Merry Gallagher, Maine DIFW.

Unit: %

Sort Order: Descending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent forest cover in the floodplain of total functional network

GIS Name: floodplain_Tot_Perc_Statewide_forestMerged

Category (not displayed in tool): Landscape

Percent forested landcover in the floodplain of a barrier's total functional network (upstream + downstream).

Forested landcover classes were aggregated separately for northern Maine and southern Maine. In northern Maine, forest cover was defined as deciduous forest, evergreen forest, mixed forest, and woody wetlands. In southern Maine, scrub/shrub was also included as a forest class based on input from Merry Gallagher, Maine DIFW.

Unit: %

Sort Order: Descending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent medium/high intensity developed landcover in the floodplain of upstream network

GIS Name: floodplain_US_Perc_nlcd_2016_developedMedHigh

Category (not displayed in tool): Landscape

Percent of medium or high intensity developed landcover in the floodplain of a barrier's upstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent medium/high intensity developed landcover in the floodplain of downstream network

GIS Name: floodplain_DS_Perc_nlcd_2016_developedMedHigh

Category (not displayed in tool): Landscape

Percent of medium or high intensity developed landcover in the floodplain of a barrier's downstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent medium/high intensity developed landcover in the floodplain of total functional network

GIS Name: floodplain_Tot_Perc_nlcd_2016_developedMedHigh

Category (not displayed in tool): Landscape

Percent of medium or high intensity developed landcover in the floodplain of a barrier's total functional network (upstream + downstream).

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent cultivated crops in the floodplain of upstream network

GIS Name: floodplain_US_Perc_nlcd_2016_cultivatedCrops

Category (not displayed in tool): Landscape

Percent of cultivated crop landcover in the floodplain of a barrier's upstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent cultivated crops in the floodplain of downstream network

GIS Name: floodplain_DS_Perc_nlcd_2016_cultivatedCrops

Category (not displayed in tool): Landscape

Percent of cultivated crop landcover in the floodplain of a barrier's downstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent cultivated crops in the floodplain of total functional network

GIS Name: floodplain_Tot_Perc_nlcd_2016_cultivatedCrops

Category (not displayed in tool): Landscape

Percent of cultivated crop landcover in the floodplain of a barrier's total (upstream + downstream) functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent pasture/hay in the floodplain of upstream network

GIS Name: floodplain_US_Perc_nlcd_2016_pastureHay

Category (not displayed in tool): Landscape

Percent of pasture/hay landcover in the floodplain of a barrier's upstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent pasture/hay in the floodplain of downstream network

GIS Name: floodplain_DS_Perc_nlcd_2016_pastureHay

Category (not displayed in tool): Landscape

Percent of pasture/hay landcover in the floodplain of a barrier's downstream functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

Percent pasture/hay in the floodplain of total functional network

GIS Name: floodplain_Tot_Perc_nlcd_2016_pastureHay

Category (not displayed in tool): Landscape

Percent of pasture/hay landcover in the floodplain of a barrier's total (upstream + downstream) functional network.

Unit: %

Sort Order: Ascending

Data Sources:

- Land Cover (NLCD 2021): MRLC Consortium
- Floodplains (2019): Fathom, Inc.

River Size Class

GIS Field Name: NESZCL_Int

Category (not displayed in tool): "Land"scape ('Water'scape)

River size class based on [NE Aquatic Habitat Classification](#).

Sort Order: A (used to increase priority of barriers on smaller streams)

Data Sources:

- USGS 10m resolution Digital Elevation Model-derived Flow-direction raster: TNC Maine

- 1a: Headwaters (<3.861 sq.mi.)
- 1b: Creeks ($\geq 3.861 < 38.61$ sq.mi.)
- 2: Small River ($\geq 38.61 < 200$ sq. mi.)
- 3a: Medium Tributary Rivers ($\geq 200 < 1000$ sq.mi.)
- 3b: Medium Mainstem Rivers ($\geq 1000 < 3861$ sq.
- 4: Large Rivers ($\geq 3861 < 9653$ sq.mi.)
- 5: Great Rivers (≥ 9653 sq.mi.)

(measure = upstream drainage area)

Miles of tidal habitat (current) upstream of barrier

GIS Field Name: MilesUS_HATcurrent

Category (not displayed in tool): Landscape/Fish Habitat

Miles of tidal habitat upstream of a barrier where tidal habitat is defined using the Highest Annual Tide Unit: Miles

Sort Order: Descending

Data Sources:

- Tidal Waters / Sea-level Rise Scenarios (2018): MGS

Miles of tidal habitat (current HAT +1m) upstream of barrier

GIS Field Name: MilesUS_HATPlus1m

Category (not displayed in tool): Landscape/Fish Habitat

Miles of tidal habitat upstream of a barrier where tidal habitat is defined using the Highest Annual Tide (HAT2018plus 1m dataset):

Unit: Miles

Sort Order: Descending

Data Sources:

- Tidal Waters / Sea-level Rise Scenarios (2018): MGS

Miles of tidal habitat (current HAT +6ft) upstream of barrier

GIS Field Name: MilesUS_HATPlus6ft

Category (not displayed in tool): Landscape/Fish Habitat

Miles of tidal habitat upstream of a barrier where tidal habitat is defined using the Highest Annual Tide (HAT2018) plus 6ft dataset:

Unit: Miles

Sort Order: Descending

Data Sources:

- Tidal Waters / Sea-level Rise Scenarios (2018): MGS

Probability of brook trout occupancy $\geq 50\%$ (current temps)

GIS Field Name: bkt_curr_occ_gt50

Category (not displayed in tool): Fish Habitat – Inland Brook Trout (BKT)

Occupancy probability of Eastern Brook trout for the catchment (NHDHRDV2) that the barrier is located within is $\geq 50\%$, given current conditions.

Unit: Boolean

Sort Order: Descending

Data Sources:

- EcoSheds Catchments (2024): USGS/EcoSheds program

High or Very High quality brook trout habitat upstream OR downstream of barrier

GIS Field Name: BT_1_4_HVH

Category (not displayed in tool): Fish Habitat – Inland Brook Trout (BKT)

Prioritization Scenario(s): Inland Brook Trout

Presence of ¼ mile or more of high, or very high priority Eastern Brook trout habitat in either the [Upstream Functional Network](#) or [Downstream Functional Network](#)

Unit: Boolean

Sort Order: Descending

Data Sources:

- Brook Trout Habitat Rating (2024): IFW

Very High-quality brook trout habitat upstream OR downstream of barrier

GIS Field Name: BT_1_4_VH

Category (not displayed in tool): Fish Habitat – Inland Brook Trout (BKT)

Presence of ¼ mile or more of very high priority Eastern Brook trout habitat in either the [Upstream Functional Network](#) or [Downstream Functional Network](#)

Unit: Boolean

Sort Order: Descending

Data Sources:

- Brook Trout Habitat Rating (2024): IFW

Probability of brook trout occupancy (current temps)

GIS Field Name: occ_current

Category (not displayed in tool): Fish Habitat – Inland Brook Trout (BKT)

Occupancy probability of Eastern Brook trout for the catchment (NHDHRDV2) that the barrier is located within, given current conditions.

Unit: %

Sort Order: Descending

Data Sources:

- EcoSheds Catchments (2024): USGS/EcoSheds program

Probability of brook trout occupancy (+4°C)

GIS Field Name: occ_temp7p40

Category (not displayed in tool): Fish Habitat – Inland Brook Trout (BKT)

Occupancy probability of Eastern Brook trout for the catchment (NHDHRDV2) that the barrier is located within, given a 4°C increase in July water temperature.

Unit: %

Sort Order: Descending

Data Sources:

- EcoSheds Catchments (2024): USGS/EcoSheds program

Probability of brook trout occupancy $\geq 50\%$ (+4°C)

GIS Field Name: bkt_p40_occ_gt50

Category (not displayed in tool): Fish Habitat – Inland Brook Trout (BKT)

Occupancy probability of Eastern Brook trout for the catchment (NHDHRDV2) that the barrier is located within is $\geq 50\%$, given a 4°C increase in July water temperature.

Unit: Boolean

Sort Order: Descending

Data Sources:

- EcoSheds Catchments (2024): USGS/EcoSheds program

Summed salmon parr productivity in US Functional Network

GIS Field Name: US_SalmParrProd

Category (not displayed in tool): Fish Habitat – Atlantic Salmon

The summed Atlantic salmon parr productivity in a barrier's Upstream Functional Network.

Salmon parr productivity obtained from a GIS-based habitat model that predicts the amount of accessible Atlantic salmon rearing habitat throughout the Gulf of Maine Atlantic salmon DPS

Units: # (Predicted parr production based on habitat units and predicted parr density)

Sort Order: Descending

Data Sources:

- Atlantic Salmon Modeled Habitat Units (2012): USFWS Gulf of Maine Coastal Program, NOAA

Species of Management Concern on one side of a barrier and not the other - Confirmed only

GIS Field Name: InvasiveBarrierConfirmed

Category (not displayed in tool): Invasive Species

Species of management concern are confirmed to be found on one side of a barrier and not the other. Species evaluated include Black crappie, Largemouth bass, Muskellunge, Northern pike, Smallmouth bass, and Walleye.

Unit: Boolean

***Note!** *This dataset was not updated for the 2025 iteration of the prioritization tool.*

Sort Order: Ascending

Data Sources:

- Fish Species of Management Concern / Invasive Fish Species (2018 approx.): IFW

Number of Species of Management Concern that are blocked - Confirmed only

GIS Field Name: InvasiveBarrierConfNumSpp

Category (not displayed in tool): Invasive Species

Number of species of management concern are confirmed to be found on one side of a barrier and not the other. Species evaluated include Black crappie, Largemouth bass, Muskellunge, Northern pike, Smallmouth bass, and Walleye.

***Note!** *This dataset was not updated for the 2025 iteration of the prioritization tool.*

Unit: #

Sort Order: Ascending

Data Sources:

- Fish Species of Management Concern / Invasive Fish Species (2018 approx.): IFW

Appendix III – Maine Aquatic Barrier Prioritization Tool - Data Sources – 2026

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Select Entity Names	Entity Abbreviation
Maine Department of Inland Fisheries and Wildlife	IFW
Maine Department of Marine Resources	DMR
Maine Geological Survey	MGS
National Oceanic and Atmospheric Administration	NOAA
U.S. Department of Agriculture – Natural Resources Conservation Service	USDA – NRCS
U.S. Fish and Wildlife Service	USFWS
United State Geologic Survey	USGS
The Nature Conservancy, Maine	TNC Maine
Gulf of Maine Distinct Population Segment	GoM DPS

Data Layer Name: Alewife Ponds	
Data Description	Ponds with known spawning runs of alewife. TNC Maine worked with DMR to reconcile the data as shown in the Maine Stream Habitat viewer with data that DMR had provided TNC previously. DMR indicated (in a spreadsheet) which ponds should be added to the previous TNC dataset. This dataset is now the best available.
Data Source	DMR; TNC Maine
Date	10/2024
Date Note	Date listed is date of last edits by TNC based on information from DMR.

Data Layer Name: Aquifers	
Data Description	Dataset contains polygons of significant aquifers (glacial deposits that are a significant ground water resource) for Maine mapped at a scale of 1:24,000, from the MGS. Printed maps published by the MGS on USGS 7.5' quadrangle bases. Aquifer boundaries delineated and digitized by the MGS from data compiled on USGS 7.5' quadrangle bases. Aquifer polygons coded by yield expected from a properly constructed well (using the AQUIFERTYPE field or the field ATYPE = 0 area not mapped as aquifer; ATYPE = 1 10-50 gallons-per-minute; ATYPE = 2 greater than 50 gallons-per-minute; ATYPE = 3 island of non-aquifer material within an area mapped as aquifer).

Data Source	MGS
Date	April 22, 2020
Date Note	Date listed is date of last update by MGS

Data Layer Name:	Atlantic Salmon Critical Habitat
Data Description	Critical habitat for Atlantic salmon (GoM DPS) includes the stream channels within the designated stream reaches and includes a lateral extent as defined by the ordinary high-water line (33 CFR 329.11). In areas where the ordinary high-water line has not been defined, the lateral extent will be defined by the bankfull elevation. Bankfull elevation is the level at which water begins to leave the channel and move into the floodplain and is reached at a discharge which generally has a recurrence interval of 1 to 2 years on an annual flood series. Critical habitat in estuaries is defined by the perimeter of the water body as displayed on standard 1:24,000 scale topographic maps or the elevation of extreme high water, whichever is greater. Watershed boundaries provide a standard level of detail that is an acceptable unit to describe salmon habitat, and the watershed level 5 (HUC 10s) are nationally recognized by the USGS to spatially relate any dataset that contains these codes.
Data Source	USDA-NRCS; NOAA
Date	July 19, 2023
Date Note	Date listed is date of last edits.

Data Layer Name:	Atlantic Salmon Habitat Quality
Data Description	Stream and River flowlines (based on 1:24K National Hydrography Dataset) rated for Atlantic salmon Habitat Quality (Low, Medium, High, Very High). Flowlines may be rated highly as spawning and rearing habitat or as important migratory rivers and streams. Data were assembled by the DMR with help from TNC Maine.
Data Source	DMR; TNC Maine
Date	10/2024
Date Note	Date listed is date of last edits by TNC based on information from DMR.

Data Layer Name:	Atlantic Salmon Habitat Recovery Units (GoM Distinct Population Segment)
Data Description	Polygon of the GoM DPS for Atlantic salmon which is divided into three sections: Merrymeeting Bay, Penobscot, and Downeast salmon habitat recovery units (SHRUs). Data sourced from DMR Open Data
Data Source	USFWS, DMR
Date	October 4, 2021
Date Note	Date listed is date of last edit.

Data Layer Name:	Atlantic Salmon Modeled Habitat Units
Data Description	GIS-based habitat model that predicts the amount of accessible Atlantic salmon rearing habitat throughout the GoM DPS. The model was developed using data from habitat surveys conducted in the Machias, Sheepscot, Dennys, Sandy, Piscataquis, Mattawamkeag, and Soudabscook Rivers. The model uses reach slope derived from contour and digital elevation model (DEM) datasets, cumulative drainage area, and physiographic province to predict the total amount of rearing habitat within a reach. The variables included in the model explain 73% of the variation in rearing habitat.
Data Source	USFWS Gulf of Maine Coastal Program; NOAA
Date	April 2012
Date Note	Date model completed.

Data Layer Name:	Atlantic Salmon State/Federal Active Management Areas
Data Description	Similar to Atlantic salmon Habitat Quality. Dataset represents ranked management priority within State and Federal Active Management Areas – that is, within watersheds classified as Atlantic salmon Critical Habitat. Stream and River flowlines (based on 1:24K National Hydrography Dataset) rated for Atlantic salmon Habitat Quality (Low, Medium, High, Very High). Flowlines may be rated highly as spawning and rearing habitat or as important migratory rivers and streams.
Data Source	DMR with help from TNC Maine
Date	10/2024
Date Note	Date listed is date of last edits by TNC based on information from DMR.

Data Layer Name:	Atlantic Salmon Surveyed Spawning and Rearing Habitat
Data Description	This coverage was developed from field surveys conducted on the mainstem and selected tributaries of the Aroostook, Dennys, Ducktap, East Machias, Kennebec, Machias, Passagassawakeag, Penobscot, Pleasant, Presumpscot, Saco, Sheepscot, St. George, Tunk, and Union Rivers in Maine by staff of the DMR - Division of Sea-Run Fisheries and Habitat. These surveys were conducted to identify important Atlantic salmon habitat including spawning and rearing areas. The majority of the survey data was collected using Trimble Pro, Pro-XL and GeoExplorer3 receivers and survey files were differentially corrected to provide 2-5 meter accuracy. Surveys for some reaches were collected with minimal or no GPS control points and the attributes were overlaid on a stream centerline created using either a GPS-acquired line, a line derived from MEGIS/USGS 1:24,000 hydrography data, or a line drawn as a centerline based on MEGIS digital orthophotography. The dataset includes information on habitat categories and areas, and an indication of spawning and rearing potential. This data is referred to as Level 3, or detailed habitat survey data, to be contrasted with the Level 2 habitat data which contains the most detailed data for individual habitat units.
Data Source	DMR
Date	March 29, 2024
Date Note	Date of last update by DMR

Data Layer Name:	Brook Trout Habitat Rating (MDIFW 2024)
Data Description	This database is a synthesis of expert recommendations from fisheries biologists from both IFW Division of Fisheries & Hatcheries and DMR Division of Sea-Run Fisheries and Ecosystems. The purpose is to rank priority stream networks and individual reaches through each of the state's sub watersheds for the conservation and restoration of habitat for native fishes. The primary intended use of the Wild Brook Trout Habitat Rating effort is to identify areas for a variety of conservation actions, such as land protection strategies, stream connectivity enhancement or instream habitat improvement. Reaches were classified into one of 5 eligible categories: Very High Value, High Value, Moderate Value, Low Value, or NOT Connect. The NOT Connect category identifies stream reaches to maintain barriers to keep invasive exotic fish from accessing native species habitats in other priority waters. To date, this database has focused on identifying the best current and potential future habitat

	for Eastern brook trout, landlocked Atlantic salmon, Atlantic sea-run salmon, and sea-run rainbow smelt.
Data Source	IFW, with assistance from TNC Maine
Date	September 17, 2024
Date Note	Date listed is date of last edits. A downloaded version of the 2024 data was used for the analysis and displayed in the 'Layers' tab because the analysis is completed before the results are uploaded to the tool and cannot be interactive. This way, users can visualize what data is underpinning the results.

Data Layer Name:	Calcareous/Moderately Calcareous Bedrock
Data Description	Regional dataset of calcareous and moderately calcareous bedrock from TNC's Ecological Land Unit raster.
Data Source	The Nature Conservancy, Ecological Land Units 2006 as referenced in: (https://www.conservationgateway.org/ConservationPlanning/SettingPriorities/EcoregionalReports/Documents/NAP_sections070131.pdf)
Date	2006
Date Note	Date listed is date of Northern Appalachians/Acadian Ecoregional Conservation Assessment that utilized the Ecological Land Units

Data Layer Name:	Coarse Sediments
Data Description	Regional dataset of coarse surficial sediments from TNC's Ecological Land Unit raster.
Data Source	The Nature Conservancy, Ecological Land Units 2006 as referenced in: (https://www.conservationgateway.org/ConservationPlanning/SettingPriorities/EcoregionalReports/Documents/NAP_sections070131.pdf)
Date	2006
Date Note	Date listed is date of Northern Appalachians/Acadian Ecoregional Conservation Assessment that utilized the Ecological Land Units

Data Layer Name:	Conservation Lands/Secured Lands
Data Description	Conservation Lands / Secured Lands maintained by TNC Maine. This dataset is, in many ways, similar to the Maine Office of GIS Conservation Lands Dataset but contains additional information on projects that TNC has been involved in. In some cases, it is more up-to-date than the Office of GIS dataset, but may be missing some smaller conservation lands parcels that are present in the Office of GIS dataset. Conservation Lands included in the current analysis included those with gap status 1,2,3, and 39 – based on TNC’s modified version of USGS’s Gap Status Codes .
Data Source	TNC Maine
Date	2/6/2025
Date Note	Date listed is date of last update by TNC Maine.

Data Layer Name:	Dams
Data Description	Point dataset of dam locations in Maine maintained by TNC Maine and Alex Abbott, independent stream restoration specialist. Current dataset is a modified version of dams point feature class in the 2022 Maine stream barrier database. TNC Maine conducted an extensive QA/QC and editing effort of the 2022 dataset in late 2024 using American Rivers Dam Removal Database 2024, Atlantic States Marine Fisheries Commission Fish Passage database from DMR (primarily used for confirming and editing fish passage structure information), and Maine Department of Environmental Protection Dam Dataset 2024. The resultant dataset contains approximately 167 edits to the 2022 dam points including some additions of dams that are currently barriers and dams that were removed or modified in the past that had been removed from past versions of the dataset.
Data Source	TNC Maine
Date	01/10/2025
Date Note	Date listed is date of last publication of updated dataset from Alex Abbott after incorporation of TNC 2024 edits.

Data Layer Name:	Eastern Brook Trout Joint Venture Wild Trout Patches
Data Description	Allopatric and Sympatric brook trout patches. A “patch” is defined as a group of contiguous catchments occupied by wild trout (Hudy et. al. 2013). Patches are not connected physically (i.e., they are separated by a dam, unoccupied warm water habitat, downstream invasive species, etc.) and are

	generally assumed to be genetically isolated. Allopatric refers to eastern brook trout only in a catchment. Sympatric refers to brook trout co-residing and competing with brown and rainbow trout.
Data Source	Eastern Brook Trout Joint Venture
Date	February 2025
Date Note	Date listed is date TNC received latest (2025) data release from IFW

Data Layer Name:	EcoSheds Catchments
Data Description	The EcoSheds brook trout occupancy model predicts probability of brook trout occupancy for catchments smaller than 200 sq. km in the northeastern U.S. from Maine to Virginia under various future temperature scenarios.
Data Source	USGS/EcoSheds
Date	June 11, 2024
Date Note	Date listed is date TNC downloaded latest catchments and .csv model results from ecosheds.

Data Layer Name:	Fish Species of Management Concern / Invasive Fish Species
Data Description	River and lakes with suspected and confirmed occurrences of species of management concern including black crappie, largemouth bass, muskellunge, northern pike, smallmouth bass and walleye. Data was used in the form of one feature class for each species.
Data Source	IFW
Date	2018 (Approximate)
Date Note	Date listed is approximate date last updated by IFW

Data Layer Name:	Floodplains
Data Description	100-yr floodplain obtained from FATHOM Fluvial & Pluvial models designed to approximate the riparian zone within which various landcover and other metrics were generated
Data Source	Fathom, Inc.
Date	2019
Date Note	Date listed is date TNC Maine acquired data from Fathom

Data Layer Name:	Hydrography
Data Description	Modified version of 1:24,000 High Resolution National Hydrography Dataset (NHD) flowline dataset. Modifications made by TNC Maine and Alex Abbott, Independent Stream Restoration Specialist. Modifications include edits to construct a single-flowline dendritic network, addition of Canadian flowline data for rivers downstream of Maine, and edits to specific sections of flowlines based on field visits, aerial photography, and LiDAR-based Digital Elevation Models.
Data Source	USGS; TNC Maine
Date	2023
Date Note	Date listed is year of last known modifications made to Hydrography dataset. No updates were made in 2024/2025.

Data Layer Name:	Inland Wading Bird and Waterfowl Habitat
Data Description	This dataset represents Inland Waterfowl and Wading bird Habitat (IWWH), a Significant Wildlife Habitat defined under Maine's Natural Resources Protection Act (NRPA). This dataset was developed in accordance with Maine's NRPA. Under this Act, IFW is designated as the authority for determining Significant Wildlife Habitats (SWHs). This dataset represents IWWH. Polygons with a high or moderate rating meet the Significant Wildlife Habitat definition and are protected under NRPA; low-rated polygons do not meet the definition and are not protected under NRPA. COMPREHENSIVE REVIEW OF ORGANIZED TOWNS: Boundaries and ratings of all IWWHs in Maine's organized townships were updated in 2008. These updates were based on interpretation of high-resolution aerial imagery captured between 2001 and 2007. FIELD ASSESSMENT OF LOW/MODERATE IWWHs: Questionable wetlands that scored near the low/moderate rating threshold are being field assessed to confirm or, if necessary, modify their ratings. These field assessments were conducted in MDIFW Region A (southern/western Maine) in 2014 and 2015; the updates were published in this layer on June 13, 2016. Assessments for MDIFW Region B (western/mid-coast Maine) were conducted in 2016 and 2017. Some of those updates (approximately 40 IWWHs) were published on Oct 2, 2017; the rest were published on Dec 14, 2017. Remapping for Region C began in 2018. This version (Mar 16, 2018) includes updating mapping for the towns of Amherst, Aurora, Blue Hill, Brewer, Bucksport, Clifton, Dedham, Eastbrook, Eddington, Ellsworth, Fletchers Landing, Great Pond, Holden,

	Mariaville, Orland, Orrington, Osborn, Otis, Penobscot, Surry, Verona Island, and Waltham. COMPREHENSIVE REVIEW OF UNORGANIZED TOWNS: Boundaries and ratings of all IWWHs attributes in Maine's unorganized townships are being updated. This update began in 2016 and is ongoing. It is based on interpretation of high-resolution aerial imagery captured primarily between 2013 and 2016. These updates are published approximately every 3 months for the townships completed during that period (typically about 5). The last such update was published on 2017-12-14.
Data Source	IFW
Date	April 2, 2018
Date Note	Date listed is date of publication.

Data Layer Name:	Land Cover (NLCD 2021)
Data Description	2021 National Land Cover Database published July 31, 2023 by a consortium of federal agencies. Resolution: 30m, Coverage: Conterminous United States, Classification: 20 Classes. For this analysis, a Maine (and immediate surroundings) version was extracted and various classes were grouped for summarization for various metrics (e.g. agricultural summarizations include both hay/pasture and cultivated crops).
Data Source	Multi-Resolution Land Characteristics (MRLC) Consortium
Date	7/31/2023
Date Note	Date listed is date of publication. On-the-ground condition date is 2021.

Data Layer Name:	Large Land Ownership
Data Description	Large Landownerships of Maine. Originally published by JW Sewall Co., used in-part as method for road-stream crossing ownership classification.
Data Source	Originally published by JW Sewall Co. 2018, internal edits made by TNC Maine and Forest Society of Maine.
Date	2018
Date Note	Date listed is date of original publication.

Data Layer Name:	Maine Heritage Fish Waters
Data Description	These data represent the list of designated State Heritage Fish Waters. State Heritage Fish Waters are lakes and ponds that contain State Heritage Fish, identified as Eastern Brook Trout and Arctic Charr, that have never been stocked according to any reliable records or that according to reliable records have not been stocked for at least 25 years. The list of State Heritage Fish Waters may be amended by rule and is generally reviewed and updated annually. Updated:M. Lubejko on 11.26.18: merged Tim P (2362) and Mud Pond (2360) into one polygon "TIM POND (INCLUDING MUD POND)" and swapped the locations of Lambert Pond (9775) and Unnamed Pond (9371). M. Lubejko on 1.7.19: Reduced the area of 4088 (a HW) by clipping out the boundaries of 7748 (not a HW).M. Gallagher on 12/01/2019: Added Blue Pond (1468), Martin Lake (1858), Elm Pond, Little (2444), Tobey Pond (4078), Tory Hill Pond (2334). Removed Line Pond (5162), Muscalsea Pond, Little (4034), Roberts Pond (5164). M. Gallagher on 1/3/2022: Removed Butcher Lake (1322).
Data Source	IFW
Date	February 11, 2025
Date Note	Date listed is date of analysis run using live feature service from Maine Office of GIS/IFW.

Data Layer Name:	Natural Barriers
Data Description	Point dataset of potential natural barriers to fish passage from the Maine stream barrier database maintained by TNC Maine and Alex Abbott, independent stream restoration specialist. Originally sourced from multiple datasets, many of the natural barriers were surveyed as part of the original road-stream crossing surveys conducted 2007-2019. The dataset is currently maintained as part of the Maine stream barrier database. Updates since 2019 have been sporadic.
Data Source	TNC Maine
Date	01/10/2025
Date Note	Date listed is date of last publication of updated dataset from Alex Abbott after incorporation of TNC 2024 edits.

Data Layer Name:	Road-Stream Crossings
Data Description	Point dataset of road-stream crossings from the Maine stream barrier database maintained by TNC Maine and Alex Abbott, independent stream restoration specialist. Original surveys were conducted 2007-2019. Updates since 2019 have been sporadic.
Data Source	TNC Maine
Date	01/10/2025
Date Note	Date listed is date of last publication of updated dataset from Alex Abbott.

Data Layer Name:	Sea-Run Brook Trout Streams
Data Description	Likely Habitat for Sea-Run Brook Trout (1:24,000 flowlines). In the current and previous iteration of the Aquatic Barrier Prioritization Tool, the team at TNC Maine was advised to create a new Sea-Run Brook Trout dataset using the Coastal Streams layer created by TNC Maine in combination with the IFW Brook Trout Habitat data listed above. Coastal Streams (created using the TNC method, see 'Glossary of Metrics') were selected where IFW Brook Trout Habitat ranking was High or Very High. The result was used as the Sea-Run Brook Trout Streams data layer.
Data Source	TNC Maine; IFW
Date	September 17, 2024
Date Note	Date listed is date of last edits to Brook Trout Habitat data.

Data Layer Name:	Smelt Reaches
Data Description	In 2024 TNC worked with various DMR point datasets to create a flowline (1:24,000) dataset of known smelt habitat. Flowlines ranked as 'High' and 'Very High' were used in the current prioritization analysis. Original DMR point data description: Points symbolize documented rainbow smelt spawning sites and waterways. Sites have been identified as historical or active spawning grounds by one of three sources: DMR list of known spawning sites written in 1971; GIS file created in 1984 from information gathered by the USFWS and DMR marking probable spawning sites and waterways used for passage for all diadromous species, attribute table allowing species specific information to be selected; and surveys conducted by Maine Marine Patrol in 2005, 2007, 2008 and 2009 in which officers visited streams identified in 1971 and 1984 as well as streams where they knew active

	spawning occurs. In 2005, 2007, 2008 and 2009, the presence of adults or the presence of eggs was used to confirm spawning at the location.
Data Source	DMR, TNC Maine
Date	June 2024
Date Note	Date listed is date TNC completed translation of existing DMR point data to prioritized flowlines for current analysis.

Data Layer Name:	Tidal Wading Bird and Waterfowl Habitat
Data Description	Tidal Waterfowl and Wading bird Habitat (TWWH) is a Significant Wildlife Habitat defined under Maine's Natural Resources Protection Act (NRPA). This dataset was developed in accordance with Maine's NRPA. Under this Act, DMR is designated as the authority for determining Significant Wildlife Habitats (SWHs). This dataset represents Tidal Waterfowl and Wading bird Habitat (TWWH). This coastal habitat that supports tidal waterfowl and wading birds is composed of four habitat components: mudflats, emergent wetlands (salt marshes), aquatic beds (eelgrass), and reefs (mussel bars). In 2016 IFW remapped the mudflat and salt marsh components from high-resolution aerial imagery and incorporated the most current mapping of eelgrass and mussel bars DMR. Rating of these habitats under NRPA is based on size of the habitat area and proximity to other tidal habitats. Contact IFW for information on a specific TWWH polygon.
Data Source	IFW
Date	April 2, 2018
Date Note	Date listed is date of publication.

Data Layer Name:	Tidal Waters / Sea-level Rise Scenarios
Data Description	This dataset approximates the potential inland extent of inundation from several scenarios (1.2, 1.6, 3.9, 6.1, 8.8 and 10.9 feet) of sea level rise or storm surge along the Maine coastline on top of the Highest Astronomical Tide. That Highest Astronomical Tide layer displays the maximum predicted astronomical high tide for the current National Tidal Datum Epoch (1983-2001). The sea level rise

	scenarios were developed by using available long-term sea level rise data from Portland, Bar Harbor, and Eastport tide gauges and the US Army Corps of Engineers Sea-Level Change Curve Calculator and sea level rise scenarios established by NOAA et al. (2017) prepared for the US National Climate Assessment. Scenarios include low, intermediate low, intermediate, intermediate high, high, and extreme sea level rise at the 50% confidence interval. The data were developed with a static (“bathtub”) inundation model that uses LiDAR topographic data as a base digital elevation model, and first adjusts Highest Astronomical Tide tidal predictions to take into account variability in elevation datums along the Maine coastline, and then adds the storm surge/sea level rise scenarios to that initial starting elevation. The primary purpose of these data is to help inform storm surge and sea level rise vulnerability assessments and community planning.
Data Source	MGS
Date	2021 (2018 scenarios)
Date Note	Date listed is date of publication by MGS

Data Layer Name:	Tribal Ownership / Trust Lands
Data Description	Tribal Lands in Maine as provided and updated by The Forest Society of Maine (FSM). TNC Maine provides additional updates.
Data Source	FSM with TNC Maine edits
Date	February 2025
Date Note	Date listed is date of last updates by FSM/TNC Maine.